

Aim

Conclusions

1. How the two methods are produced
2. The parameter sweep
3. Agreement, per run
4. The settled comparison
5. The unhashed donor
6. What was run, in order
7. Data locations
8. Tools
9. Caveats

Benchmarking: demultiplexing with genotyping vs HTO

Code ▾

2026-08-03

Runs: pilot 260423_VH01624_453_222HWMYX (where the parameter sweep was done), then 260522, 260528, 260717 through the pipeline.

Status: settled. `positive.quantile = 0.99` was adopted and is what the Nextflow pipeline runs for every sample.

Analysis finished: 2026-07-28 - the most recent of the 46 figures shown below.

Report generated: 2026-08-03 by 07.demux_benchmark_report.Rmd, from the outputs of steps 01-06 in this folder.

Samples are pooled several to a GEM well, so every cell has to be assigned back to a donor. Two independent methods are available: hashtag oligos (HTO), which are added at the bench, and genotype demultiplexing from the RNA itself via cellSnp-lite and vireo. This report compares them.

For full report, download `demux_benchmark_report.html` in the attachment and open in a browser.

Code used - attached and in the github repository: https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/Benchmarking-demultiplexing_with_genotyping_vs_HTO (https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/Benchmarking-demultiplexing_with_genotyping_vs_HTO)

Aim

Decide which demultiplexing method to carry into the pipeline, and on what setting:

- do HTO hashing and genotype demultiplexing recover the same donors
- which assigns more cells as confident singlets, and which calls more doublets
- what `positive.quantile` threshold should HTODemux use
- does the method still work for a donor that was never hashed

Conclusions

- genotyping with cellSnp/vireo outperforms HTO hashing, and is the method the pipeline now relies on for donor assignment
- HTO is still worth running: it is what names the donors, since vireo only produces anonymous `donor0...donorN` labels that have to be matched back to sample identity
- `positive.quantile = 0.99` was adopted after the sweep and is what every pipeline run uses; the pilot's original 0.80 is too permissive
- the unhashed-donor case works after the fix in `nextflow/bin/process_cellsnp_hto.R`, verified against LK3's real output rather than a reimplementation

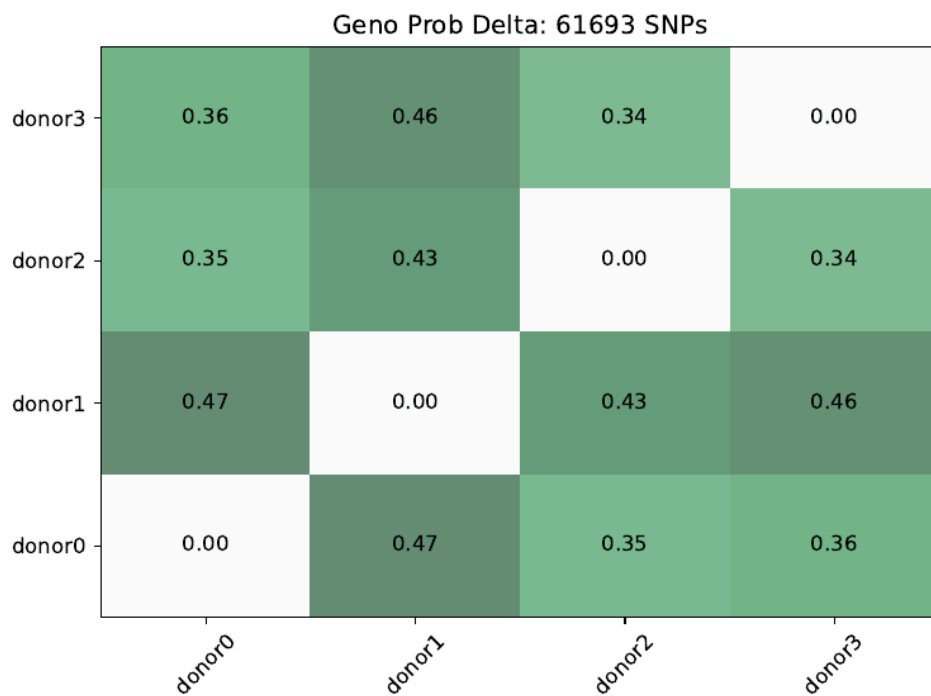
1. How the two methods are produced

HTO comes from CITE-seq-Count on the hashtag library (step 01), whitelisted to the emptyDrops barcodes, then HTODemux. Genotype demultiplexing (step 02) runs cellSnp-lite over the same barcode list against the Cell Ranger BAM, then vireo with `N_DONORS=4`. Step 03 joins the two and produces everything below.

The important asymmetry: **vireo output is anonymous**. It returns `donor0`, `donor1` ... with no idea which is which. HTO is what supplies the mapping from genotype cluster to sample name, which is why both are run even though genotyping is the better classifier. That mapping is saved as `vireo_to_hto_sample_name_map_positive_quantile_0.99.csv`.

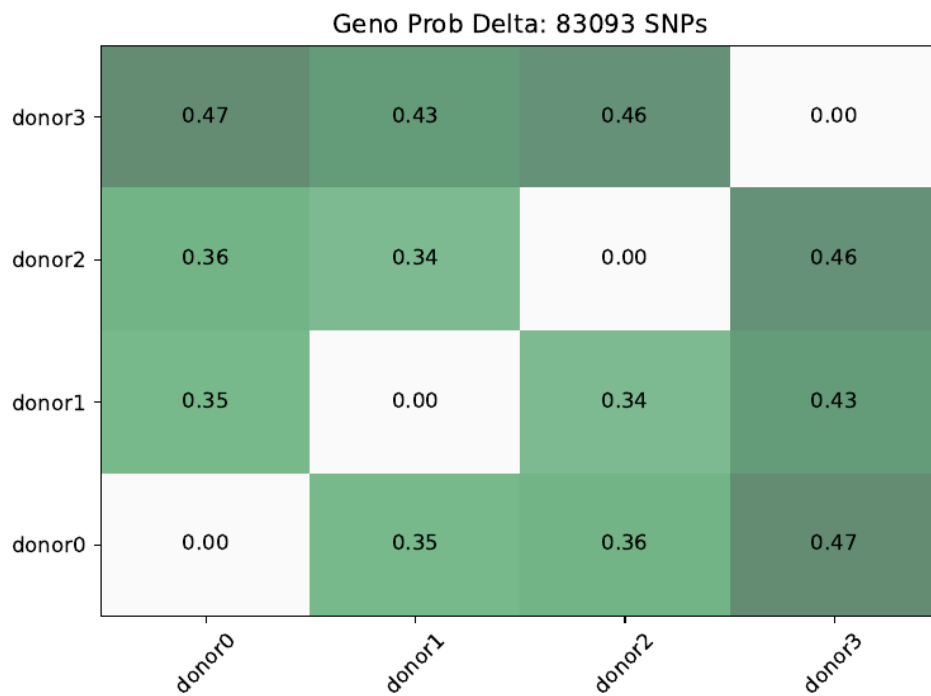
fig_GT_distance_estimated.pdf

Vireo estimated genotype distance between donors, 260423. Confirms the donors are genetically distinguishable before any comparison is made.



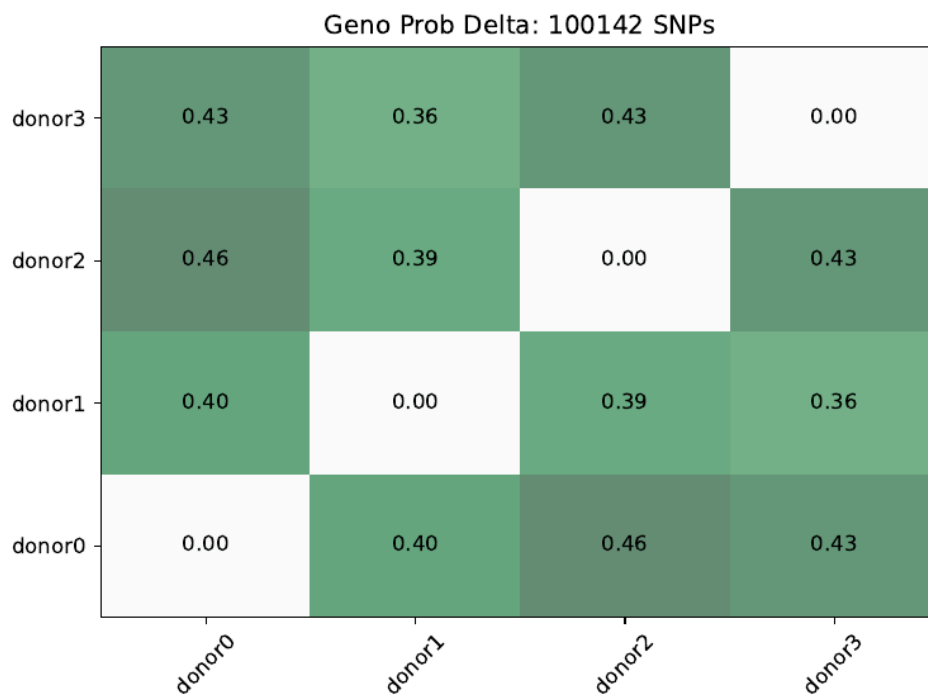
fig_GT_distance_estimated.pdf

Vireo estimated genotype distance between donors, 260522. Confirms the donors are genetically distinguishable before any comparison is made.



fig_GT_distance_estimated.pdf

Vireo estimated genotype distance between donors, 260528. Confirms the donors are genetically distinguishable before any comparison is made.



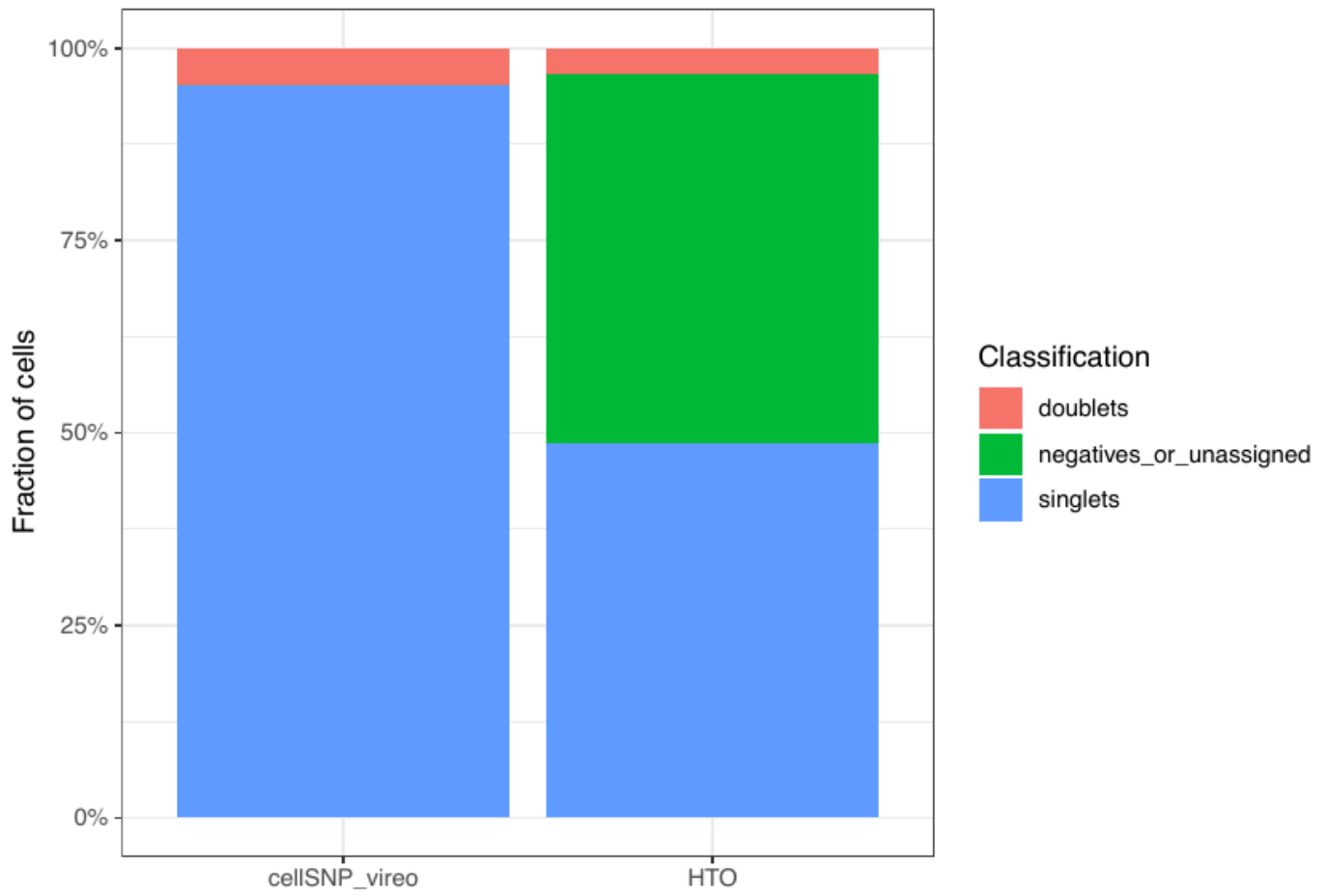
2. The parameter sweep

The sweep was done on the pilot run. HT0Demux's `positive.quantile` sets how aggressively a cell is called positive for a hashtag; every comparison figure was produced at the default 0.80, at 0.99, and across all quantiles.

01_demux_efficiency_barplot.pdf

01_demux_efficiency_barplot at the default 0.80.

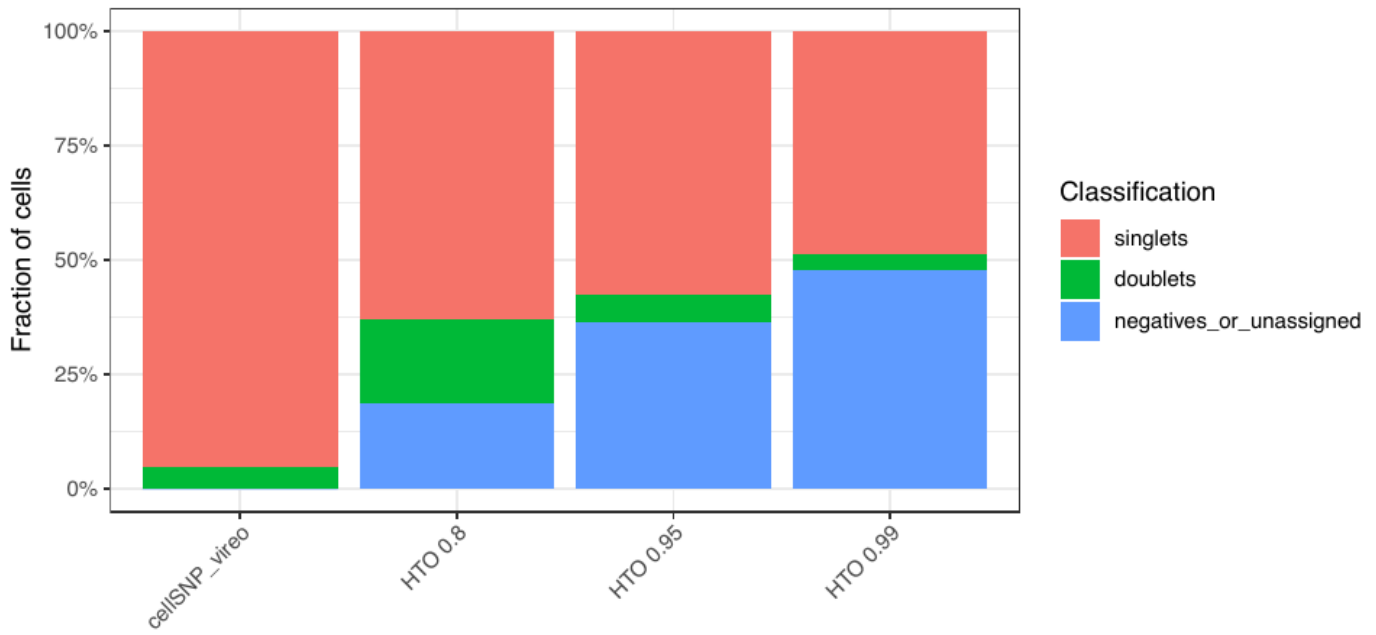
Demultiplexing efficiency: HTO vs cellSNP-vireo



01_demux_efficiency_barplot_all_positive_quantiles.pdf

01_demux_efficiency_barplot across all positive quantiles.

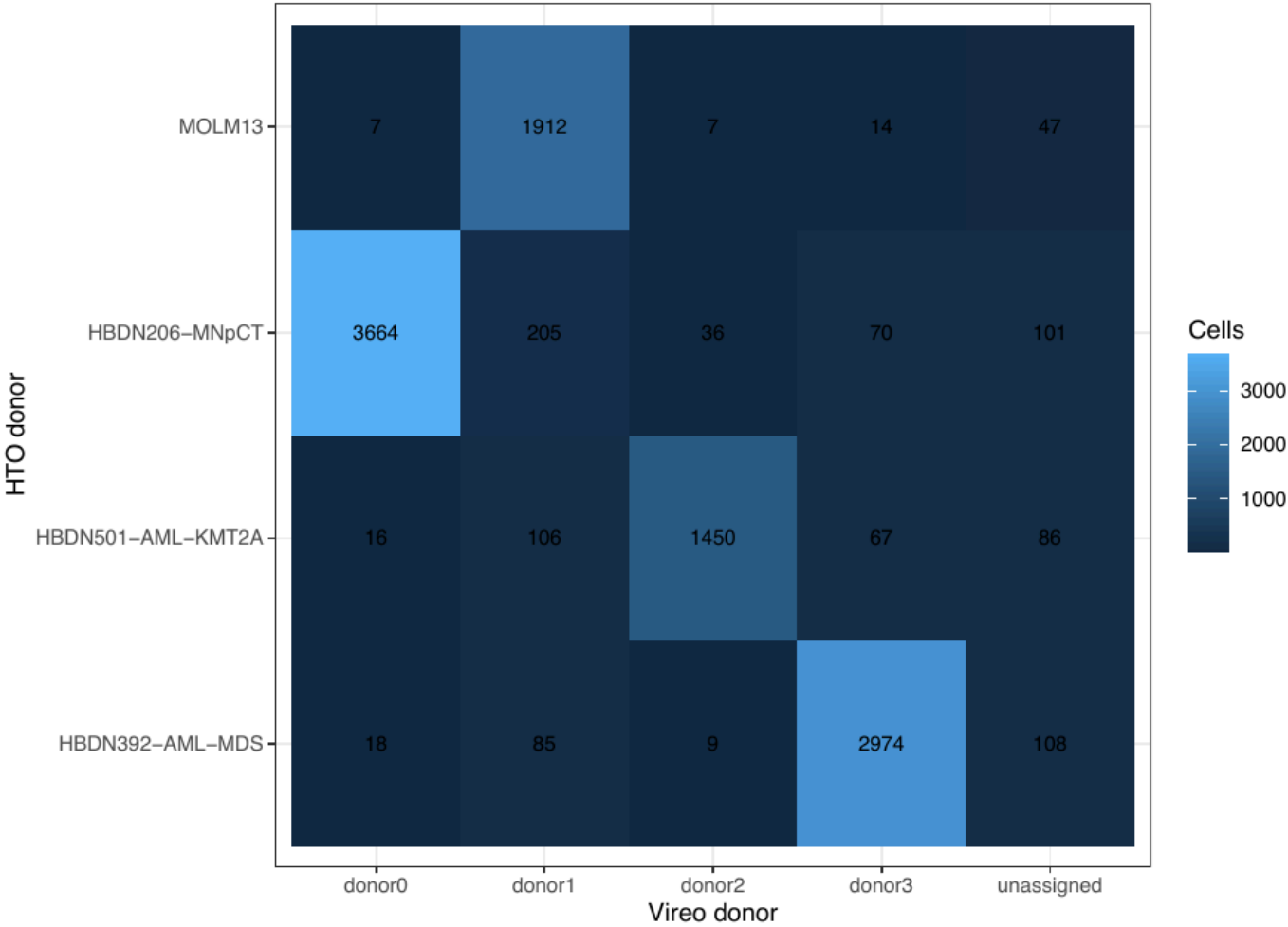
Demultiplexing efficiency: Vireo versus HTODemux thresholds



03_hto_vireo_donor_heatmap.pdf

03_hto_vireo_donor_heatmap at the default 0.80.

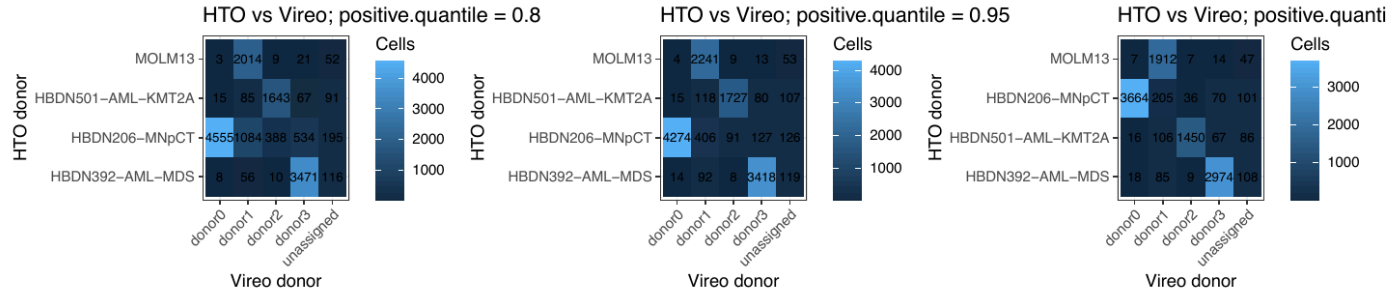
HTO versus Vireo donor assignment



03_hto_vireo_donor_heatmap_all_positive_quantiles.pdf

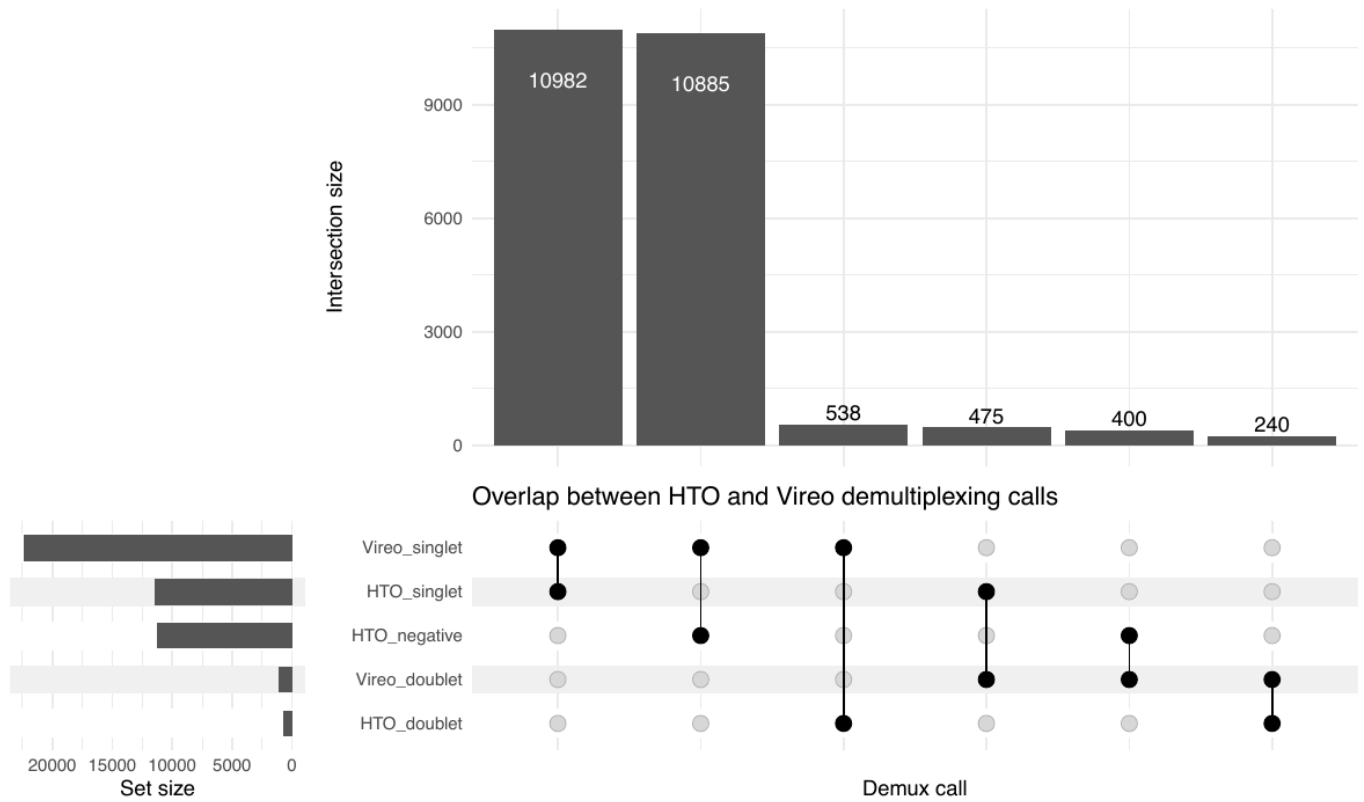
03_hto_vireo_donor_heatmap across all positive quantiles.

HTO versus Vireo donor assignment across HTODemux thresholds



04_upset_HTO_vireo_overlap.pdf

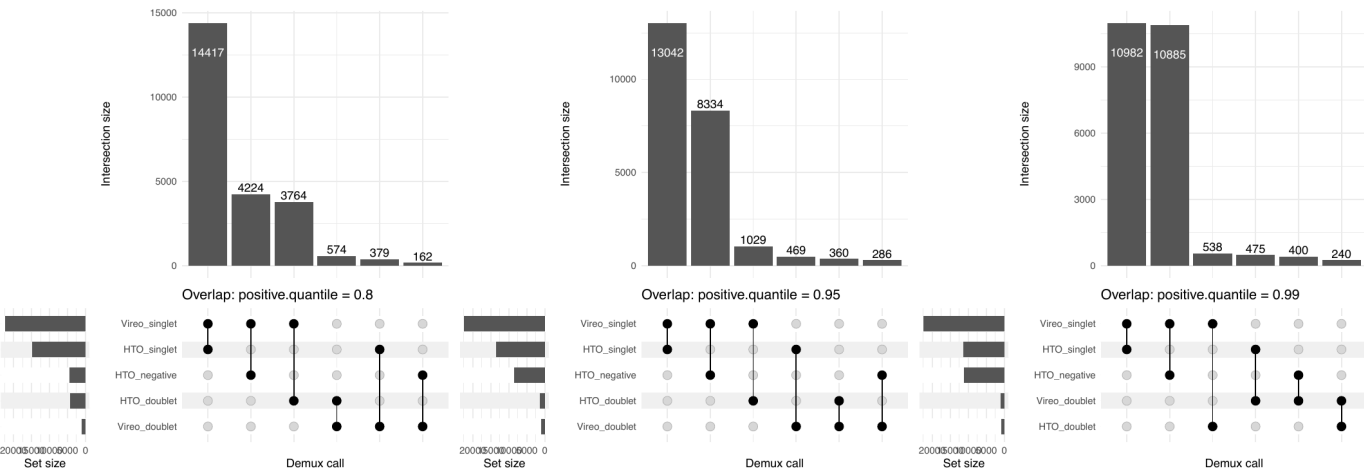
04_upset_HTO_vireo_overlap at the default 0.80.



04_upset_HTO_vireo_overlap_all_positive_quantiles.pdf

04_upset_HTO_vireo_overlap across all positive quantiles.

Overlap between HTO and Vireo calls across HTODemux thresholds



Demux efficiency, pilot run, default quantile

method	total_cells	singlets	doublets	negatives_or_unassigned	singlet_fraction	doublet_fraction	negative_or_unassigned_fraction
HTO	23520	11457	778	11285	0.4871173	0.0330782	0.4798044
cellSNP_vireo	23520	22405	1115	0	0.9525935	0.0474065	0.0000000

Demux efficiency across the quantile sweep

pq_tag	method	total_cells	singlets	doublets	negatives_or_unassigned	singlet_fraction	doublet_fraction	negative_or_unassigned_fracti
positive_quantile_0.8	HTO 0.8	23520	14796	4338	4386	0.6290816	0.1844388	0.18647
positive_quantile_0.8	cellSNP_vireo	23520	22405	1115	0	0.9525935	0.0474065	0.00000
positive_quantile_0.95	HTO 0.95	23520	13511	1389	8620	0.5744473	0.0590561	0.36649
positive_quantile_0.95	cellSNP_vireo	23520	22405	1115	0	0.9525935	0.0474065	0.00000
positive_quantile_0.99	HTO 0.99	23520	11457	778	11285	0.4871173	0.0330782	0.47980

pq_tag	method	total_cells	singlets	doublets	negatives_or_unassigned	singlet_fraction	doublet_fraction	negative_or_unassigned_fra
positive_quantile_0.99	cellSNP_vireo	23520	22405	1115	0	0.9525935	0.0474065	0.00000

Only the 0.99 outputs survive for the later runs, which is the practical record of the decision: from LK2 onwards nothing else was generated.

3. Agreement, per run

260522

From the Nextflow pipeline.

Demux efficiency - 260522

method	total_cells	singlets	doublets	negatives_or_unassigned	singlet_fraction	doublet_fraction	negative_or_unassigned_fraction
HTO	21301	12529	736	8036	0.5881883	0.0345524	0.3772593
cellSNP_vireo	21301	20996	305	0	0.9856814	0.0143186	0.0000000

HTO / vireo agreement - 260522

total_cells	both_singlet	HTO_doublet_only	Vireo_doublet_only	both_doublet	HTO_singlet_Vireo_unassigned	HTO_negative_Vireo_singlet	both_singlet_fraction
21301	12414	697	266	39	0	7885	0.58271

Vireo summary - 260522

vireo_class	vireo_donor	n_cells
singlet	donor3	5737
singlet	donor0	5681
singlet	donor2	5658
singlet	donor1	3046
singlet	unassigned	874
doublet	doublet	177
doublet	unassigned	128

260528

From the Nextflow pipeline.

Demux efficiency - 260528

method	total_cells	singlets	doublets	negatives_or_unassigned	singlet_fraction	doublet_fraction	negative_or_unassigned_fraction
HTO	33224	23897	3979	5348	0.7192692	0.1197628	0.160968
cellSNP_vireo	33224	31007	2217	0	0.9332711	0.0667289	0.000000

HTO / vireo agreement - 260528

total_cells	both_singlet	HTO_doublet_only	Vireo_doublet_only	both_doublet	HTO_singlet_Vireo_unassigned	HTO_negative_Vireo_singlet	both_singlet_fraction
33224	22910	3265	1503	714	0	4832	0.68951

Vireo summary - 260528

vireo_class	vireo_donor	n_cells
singlet	donor0	9953
singlet	donor1	9443
singlet	unassigned	5317
singlet	donor3	3994
singlet	donor2	2300
doublet	unassigned	1139

vireo_class	vireo_donor	n_cells
doublet	doublet	1078

260717

From the Nextflow pipeline.

Demux efficiency - 260717

method	total_cells	singlets	doublets	negatives_or_unassigned	singlet_fraction	doublet_fraction	negative_or_unassigned_fraction
HTO	25926	15314	1527	9085	0.5906812	0.0588984	0.3504204
cellSNP_vireo	25926	24509	1417	0	0.9453444	0.0546556	0.0000000

HTO / vireo agreement - 260717

total_cells	both_singlet	HTO_doublet_only	Vireo_doublet_only	both_doublet	HTO_singlet_Vireo_unassigned	HTO_negative_Vireo_singlet	both_singlet_fract	
25926	14652	1061	951	466		0	8796	0.565

Vireo summary - 260717

vireo_class	vireo_donor	n_cells
singlet	donor2	8391
singlet	donor0	8159
singlet	donor1	6599
singlet	unassigned	1360
doublet	doublet	975
doublet	unassigned	442

4. The settled comparison

Figures at positive.quantile = 0.99 , as the pipeline produces them.

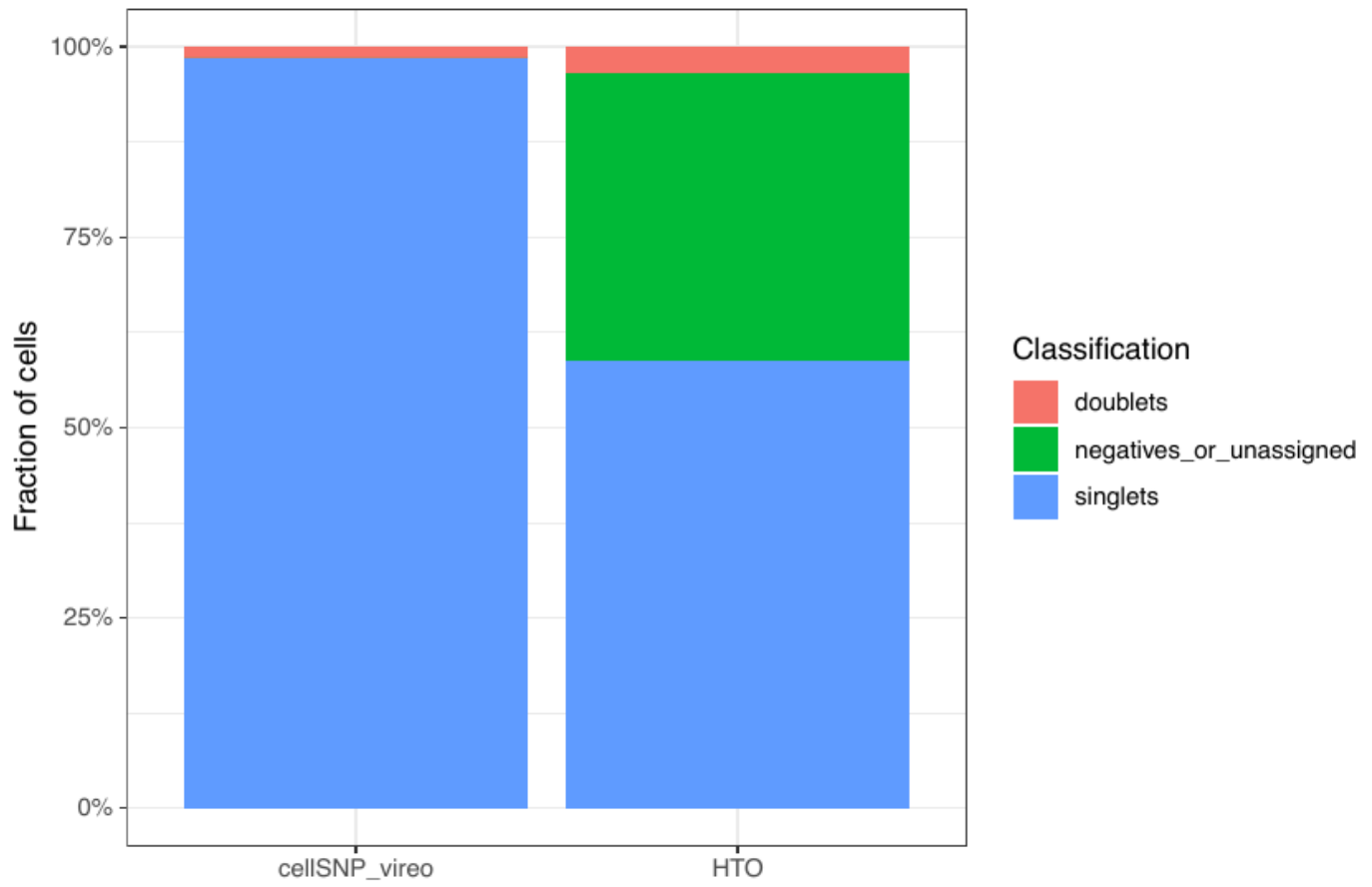
260522

01_demux_efficiency_barplot_positive_quantile_0.99.pdf

Singlet / doublet / negative rates for HTO and vireo side by side. The headline comparison.

Demultiplexing efficiency: HTO vs cellSNP-vireo

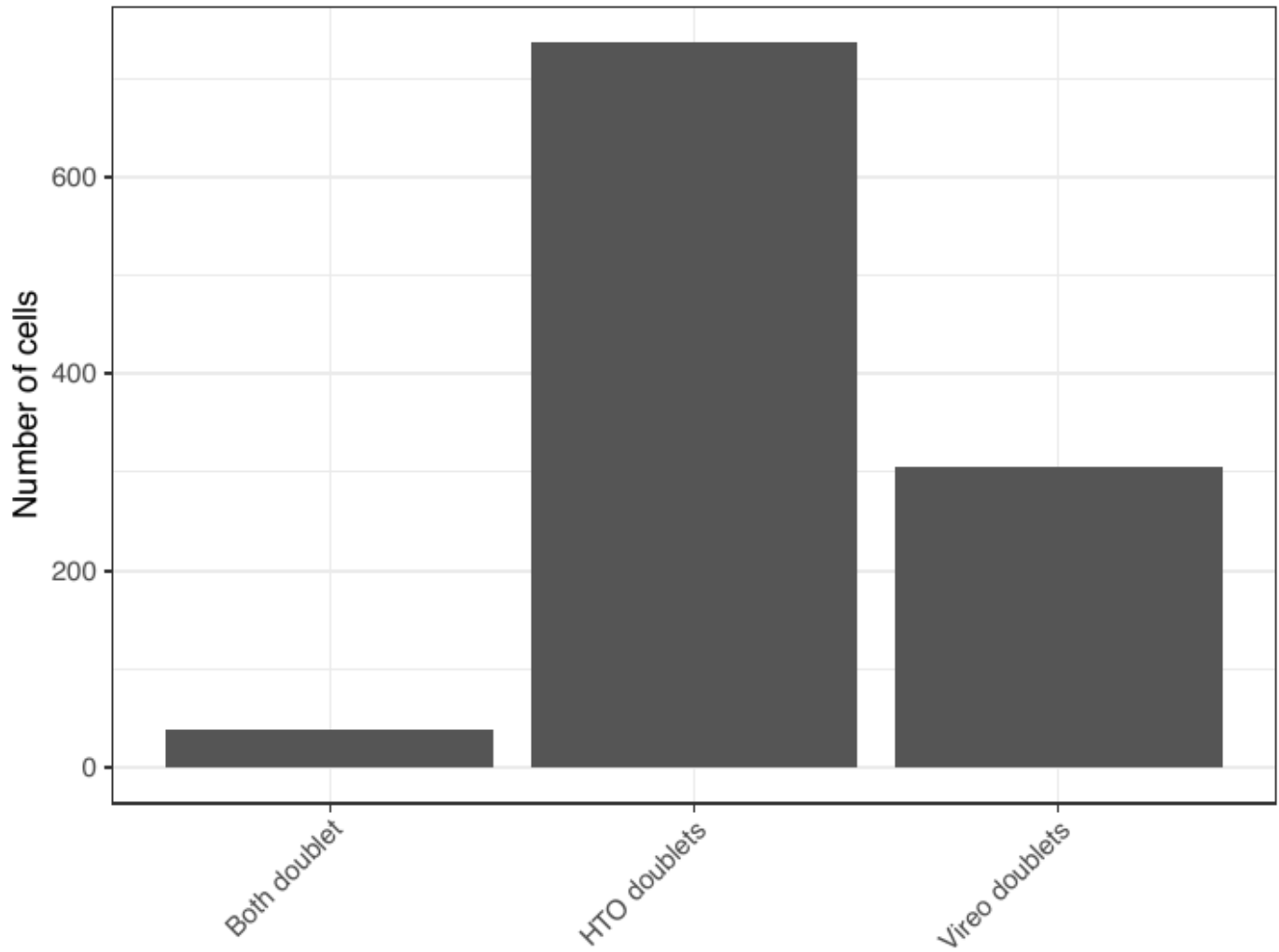
HTODemux positive.quantile = 0.99



02_hto_vireo_doublet_comparison_positive_quantile_0.99.pdf

Whether the two methods call the same cells doublets.

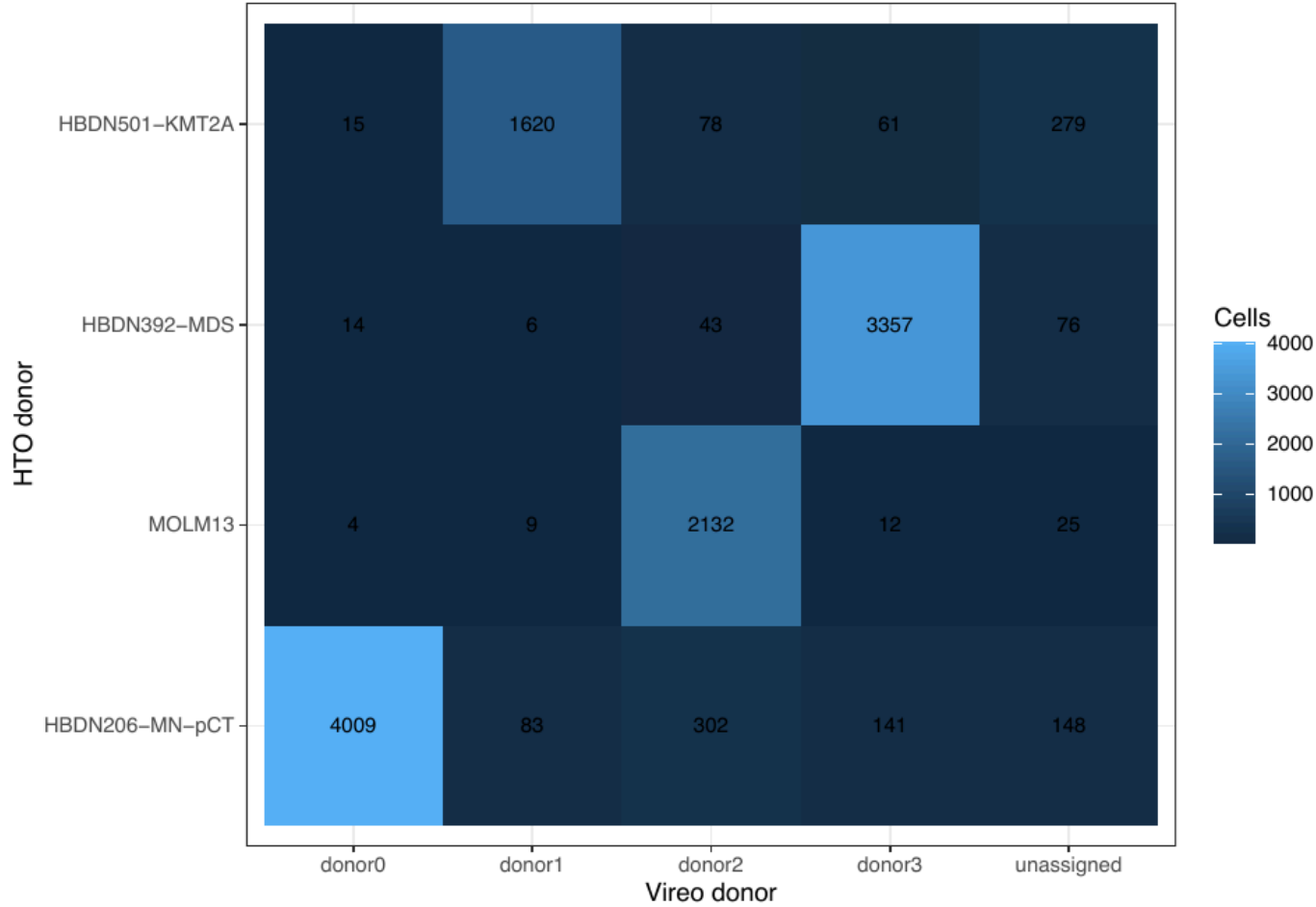
Doublet detection overlap
HTODemux positive.quantile = 0.99



03_hto_vireo_donor_heatmap_positive_quantile_0.99.pdf

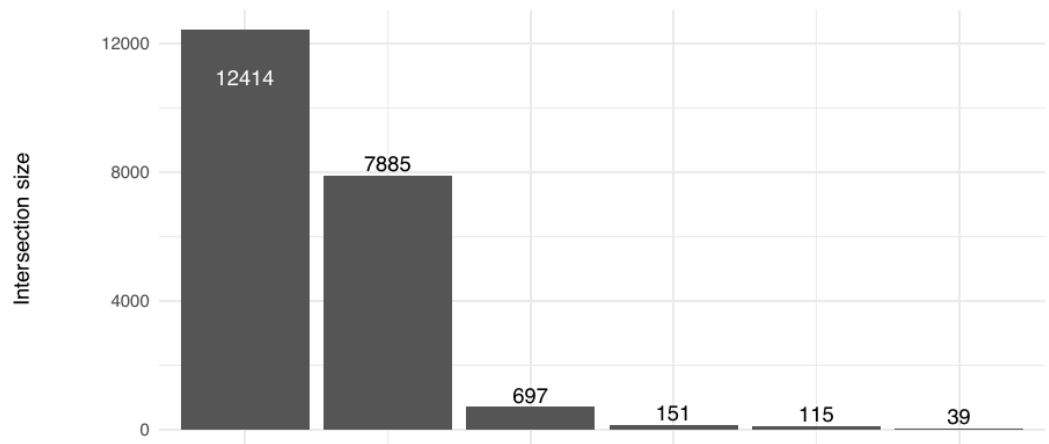
HTO sample against vireo donor. Off-diagonal mass is the disagreement; this is also the figure the sample-name mapping is read off.

HTO versus Vireo donor assignment
HTODemux positive.quantile = 0.99

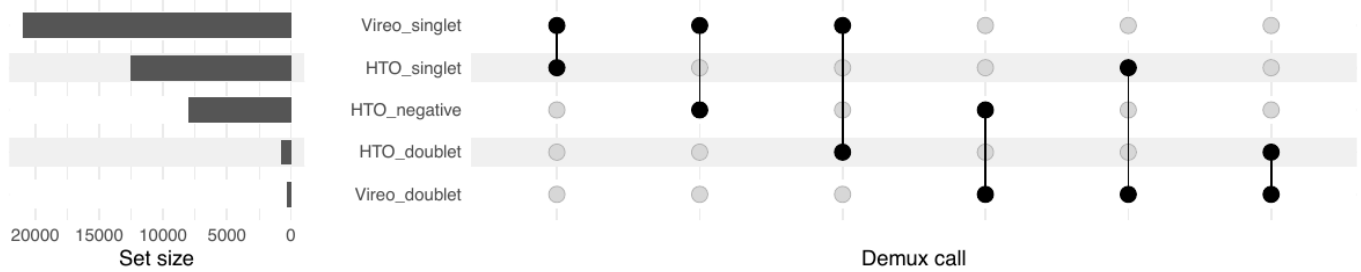


04_upset_HTO_vireo_overlap_positive_quantile_0.99.pdf

UpSet of the call overlap.



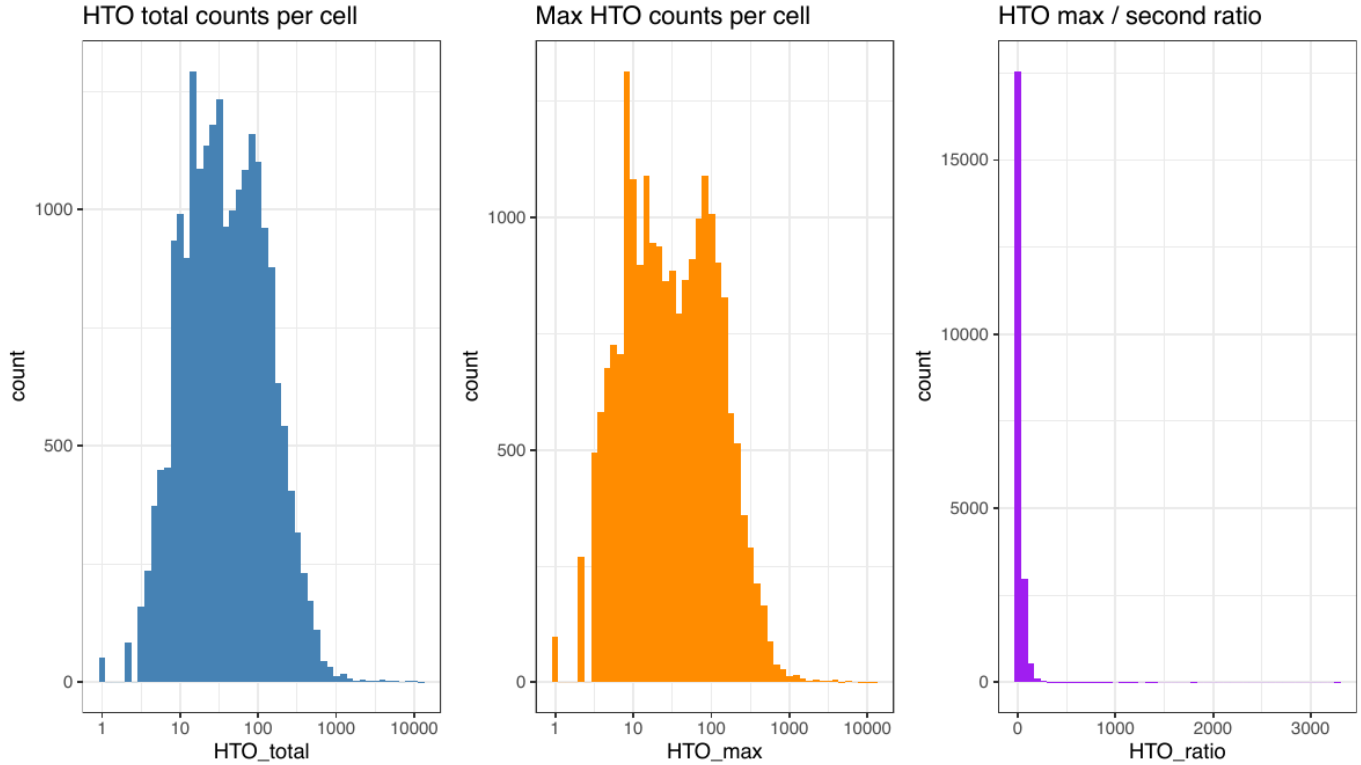
Overlap between HTO and Vireo demultiplexing calls
HTODemux positive.quantile = 0.99



05_HTO_QC_positive_quantile_0.99.pdf

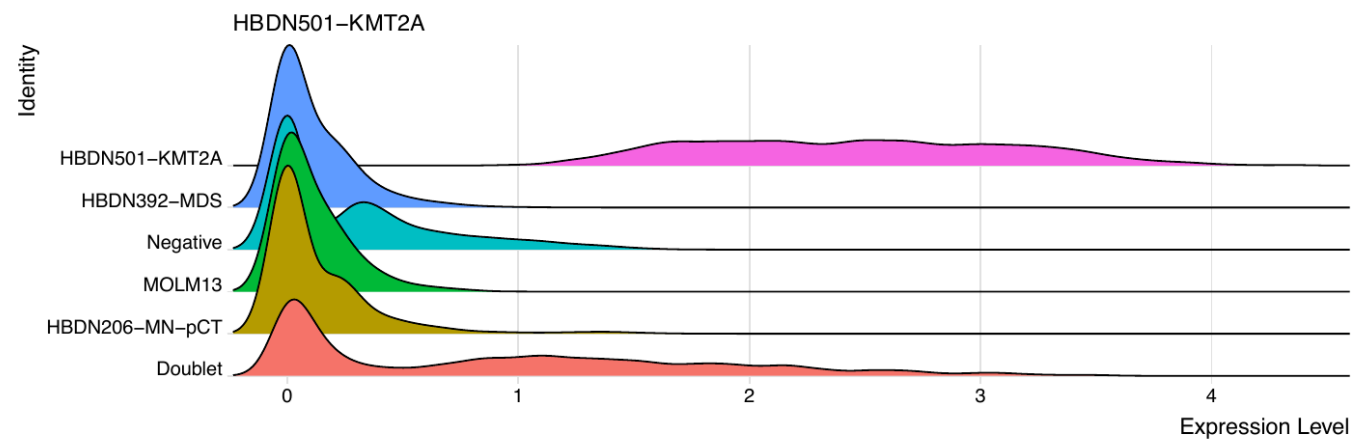
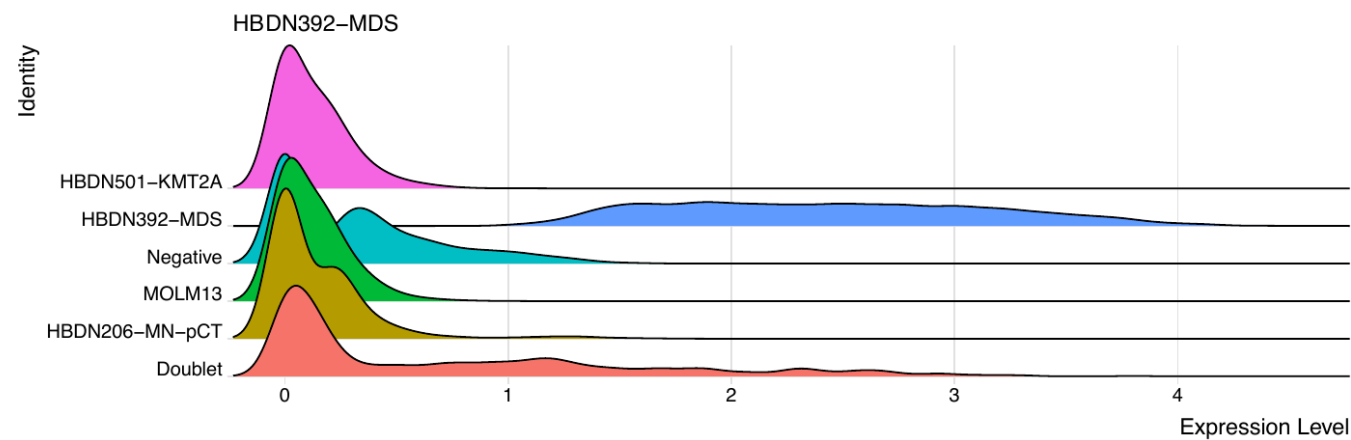
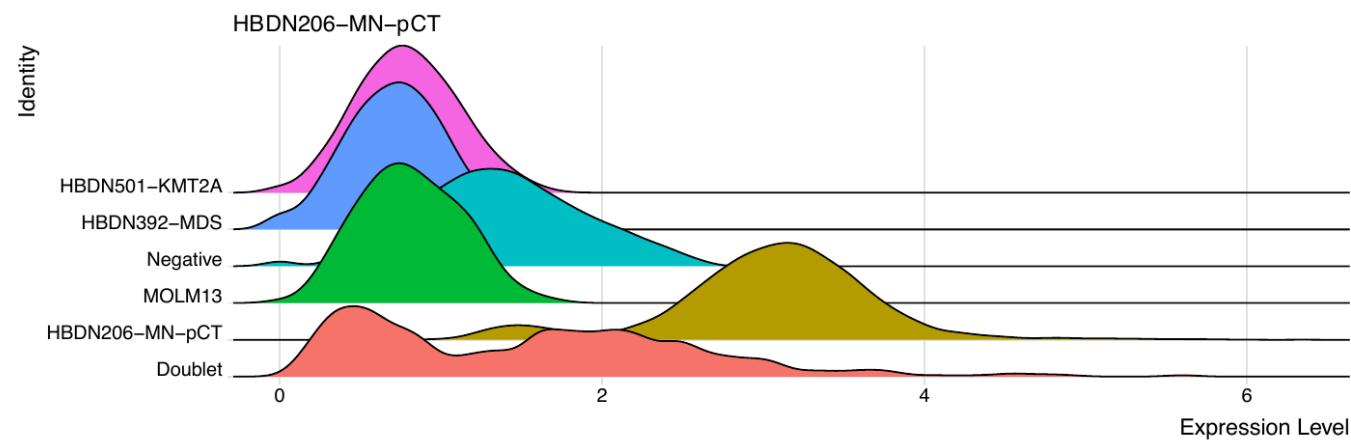
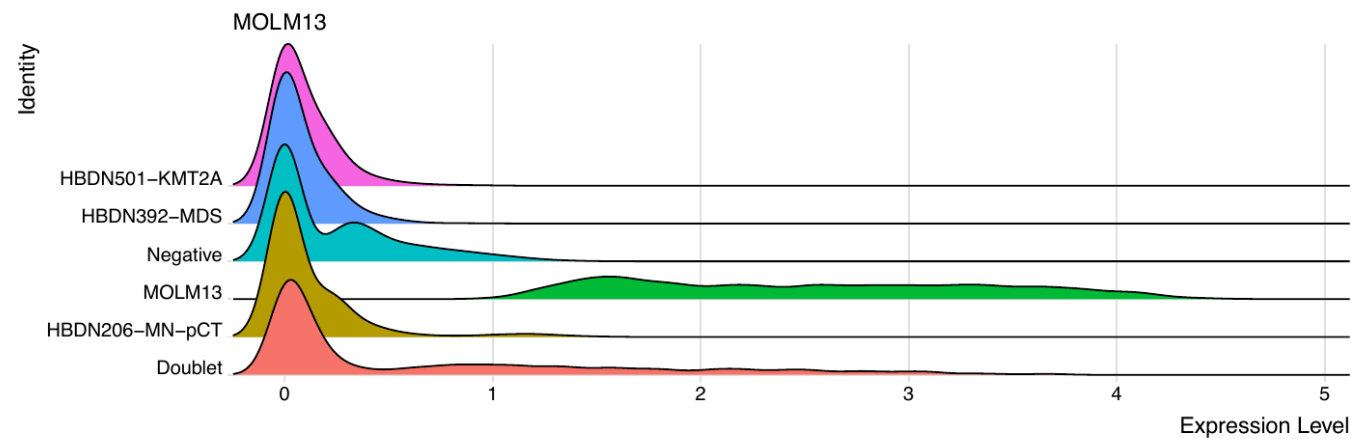
HTO QC at the adopted threshold.

HTO QC; HTODemux positive.quantile = 0.99

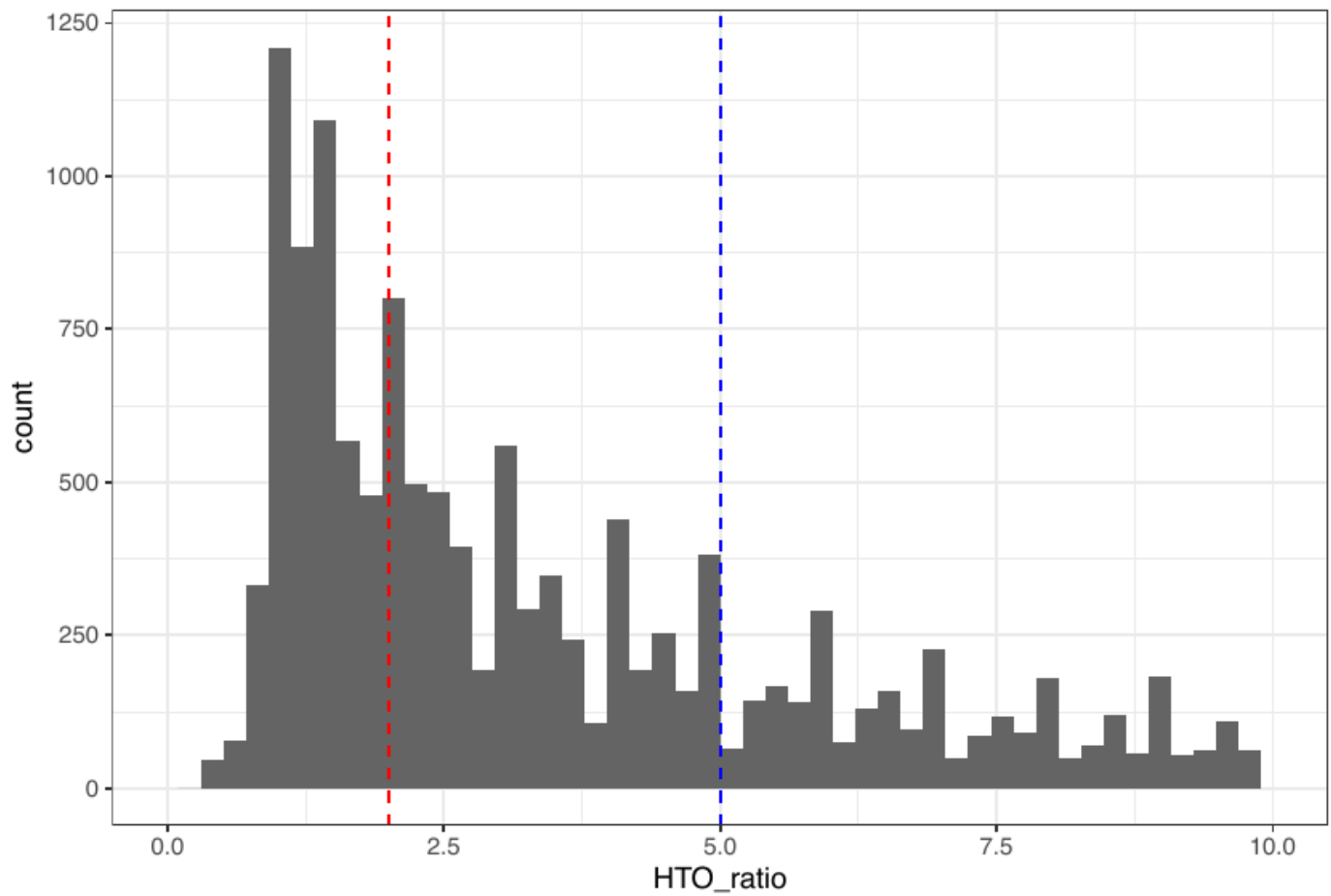


06_HTO_ridge_positive_quantile_0.99.pdf

Ridge plots per hashtag.



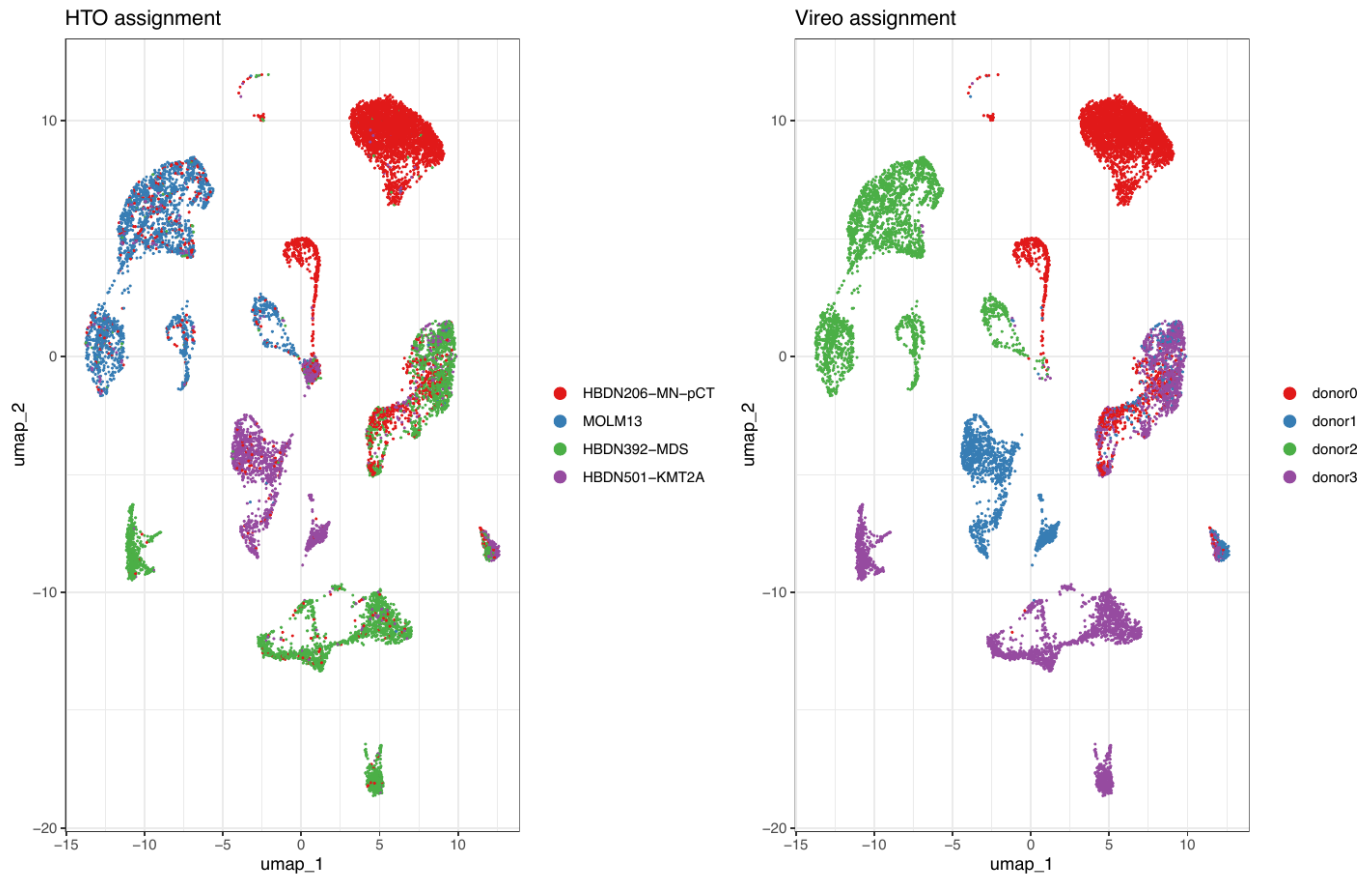
HTO ratio thresholds; HTODemux positive.quantile = 0.99



07_UMAP_HTO_cellSNP_positive_quantile_0.99.pdf

Both sets of calls on the UMAP.

HTO and Vireo assignments; HTODemux positive.quantile = 0.99



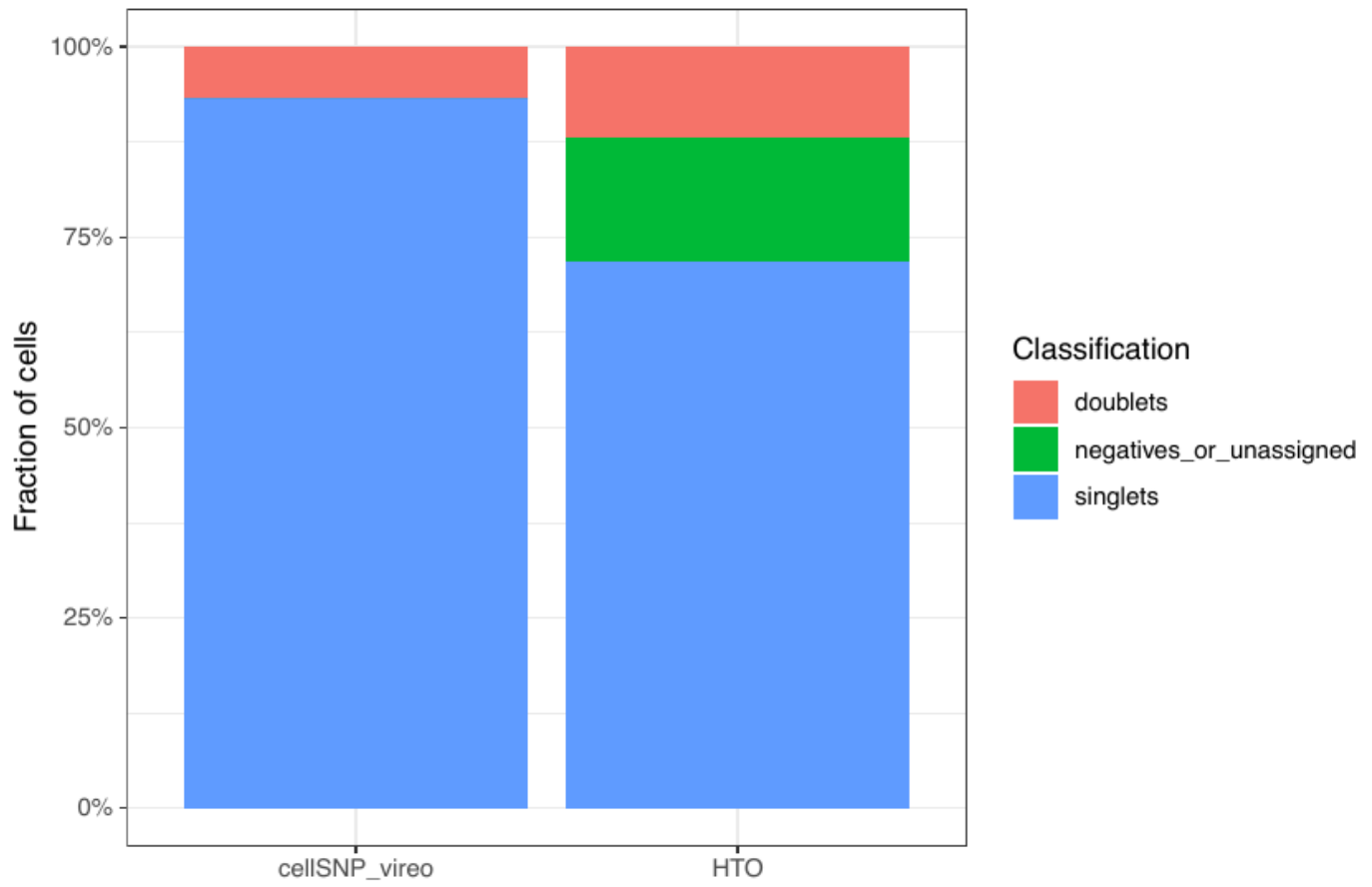
260528

01_demux_efficiency_barplot_positive_quantile_0.99.pdf

Singlet / doublet / negative rates for HTO and vireo side by side. The headline comparison.

Demultiplexing efficiency: HTO vs cellSNP-vireo

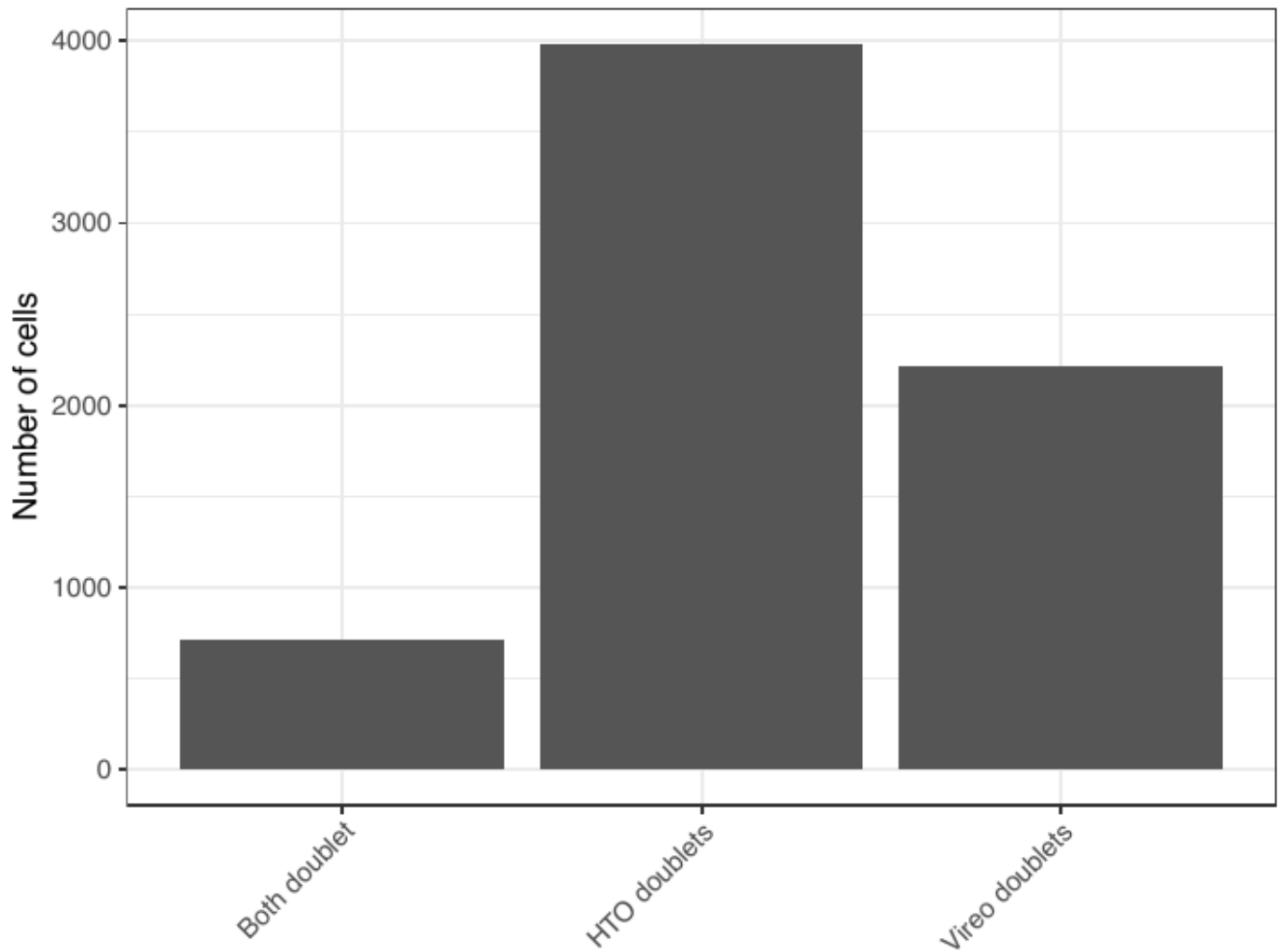
HTODemux positive.quantile = 0.99



02_hto_vireo_doublet_comparison_positive_quantile_0.99.pdf

Whether the two methods call the same cells doublets.

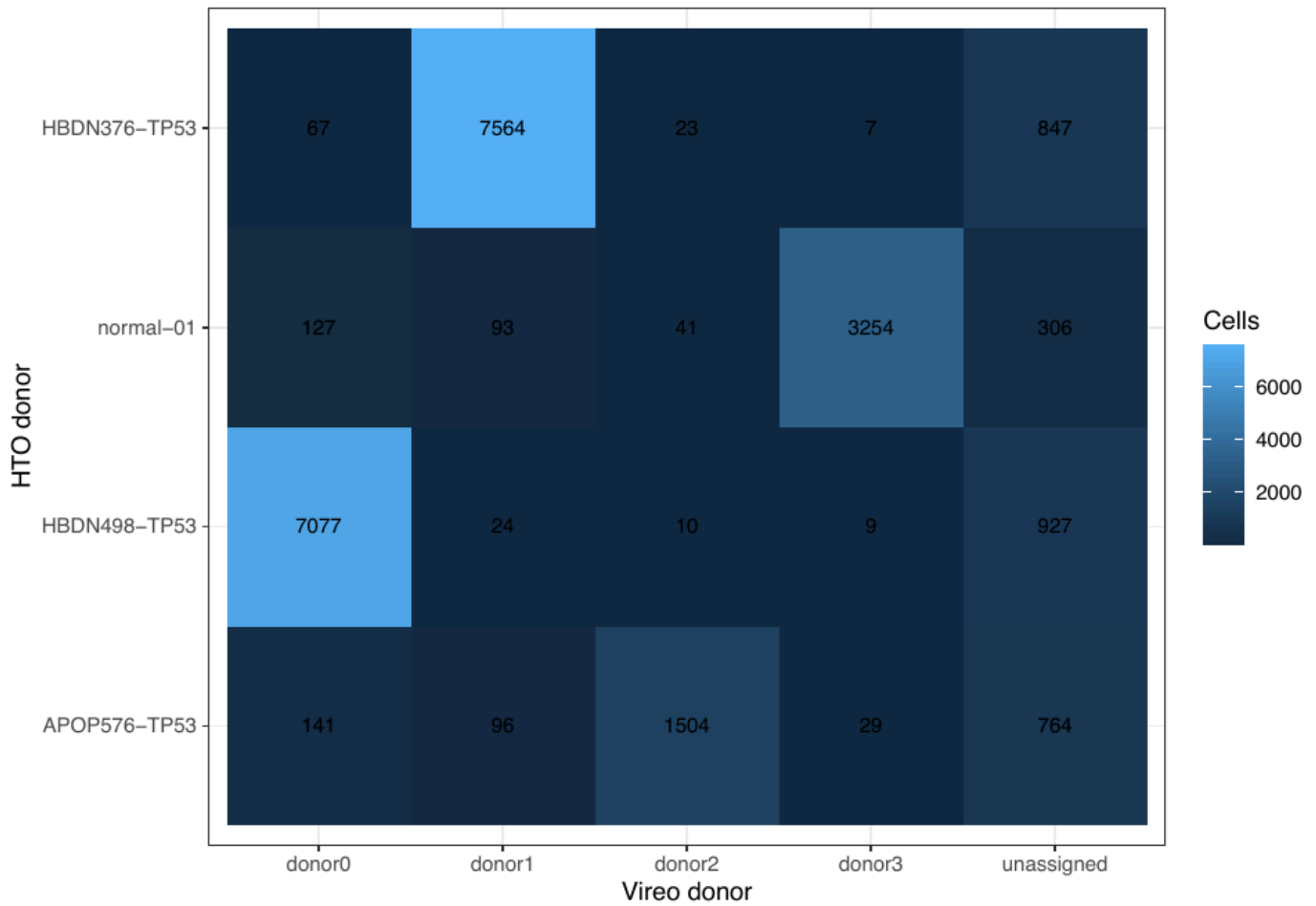
Doublet detection overlap
HTODemux positive.quantile = 0.99



03_hto_vireo_donor_heatmap_positive_quantile_0.99.pdf

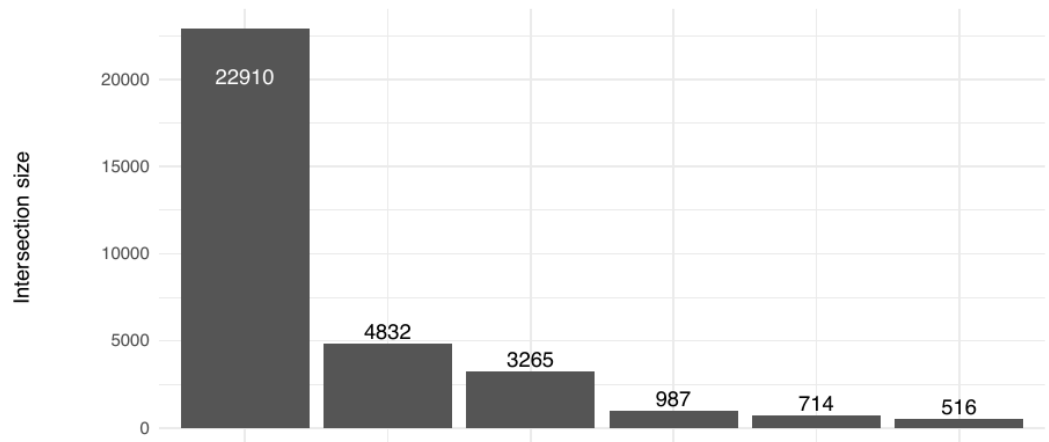
HTO sample against vireo donor. Off-diagonal mass is the disagreement; this is also the figure the sample-name mapping is read off.

HTO versus Vireo donor assignment
HTODemux positive.quantile = 0.99

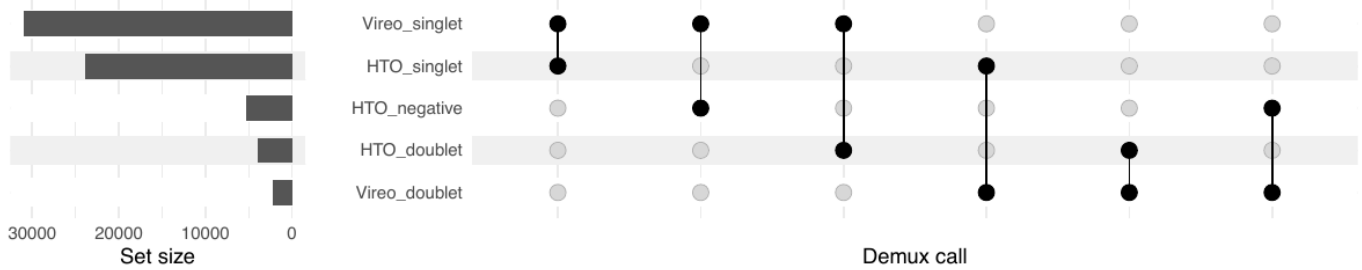


04_upset_HTO_vireo_overlap_positive_quantile_0.99.pdf

UpSet of the call overlap.



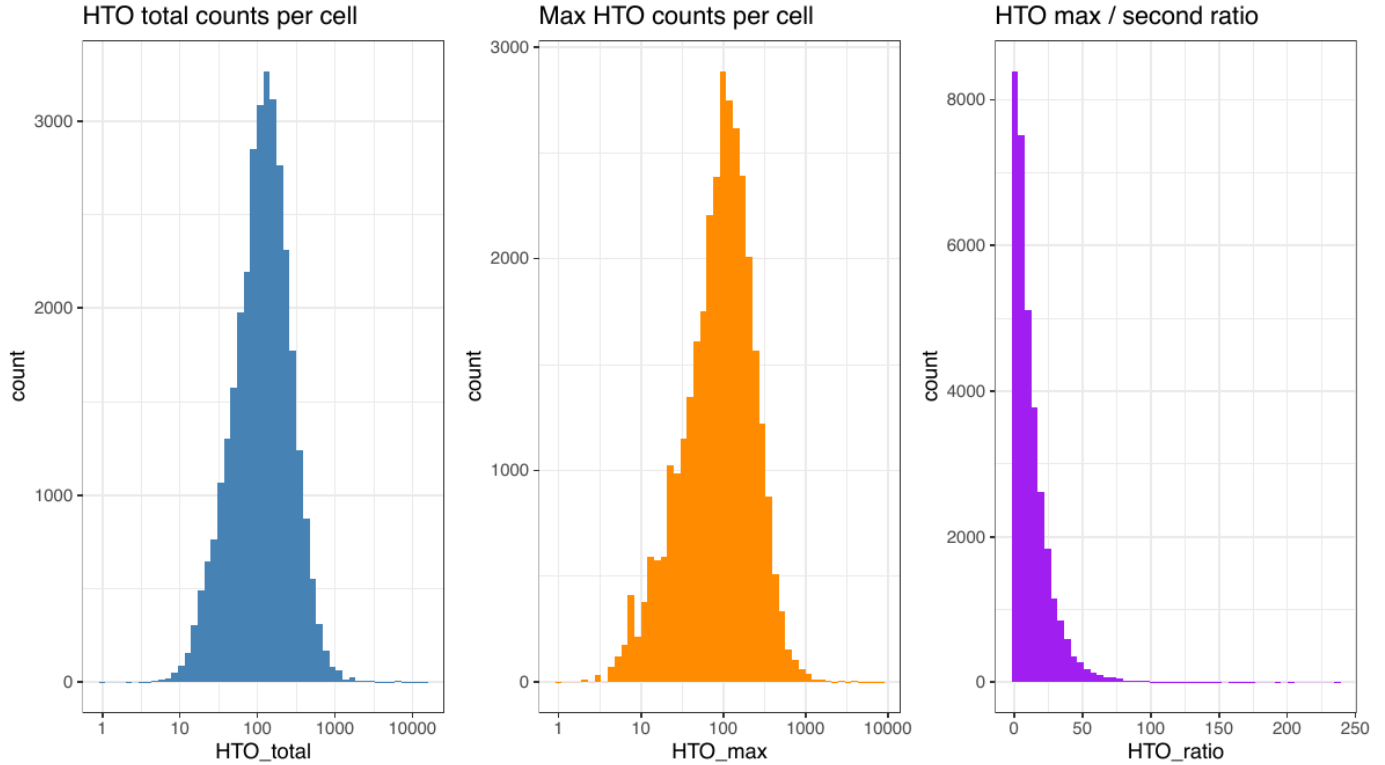
Overlap between HTO and Vireo demultiplexing calls
HTODemux positive.quantile = 0.99



05_HTO_QC_positive_quantile_0.99.pdf

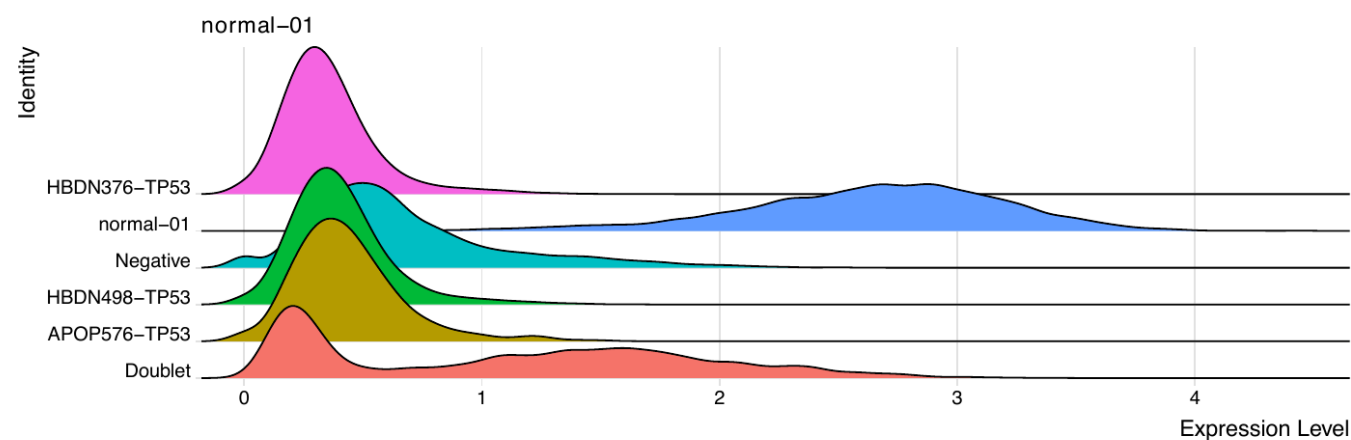
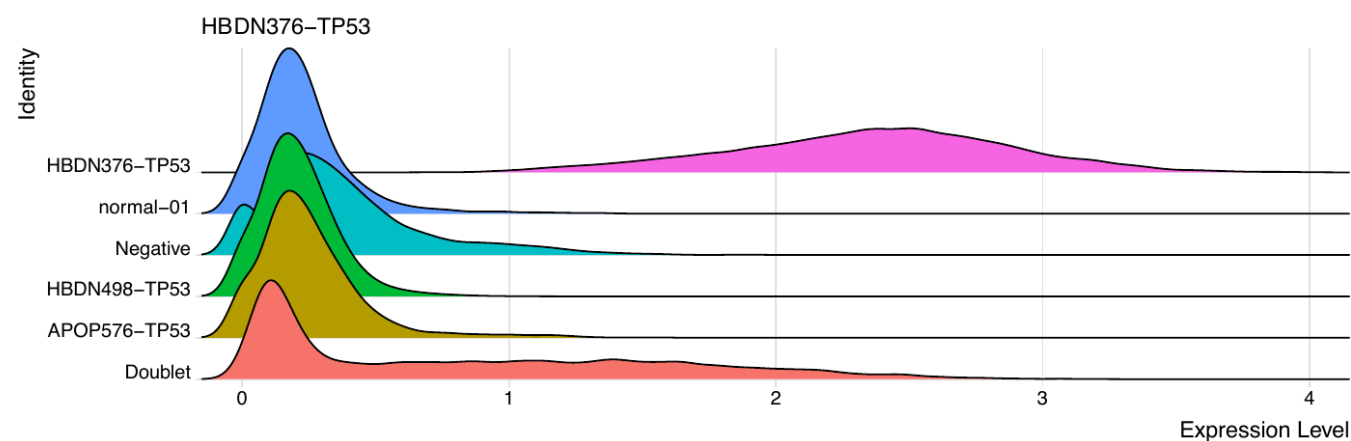
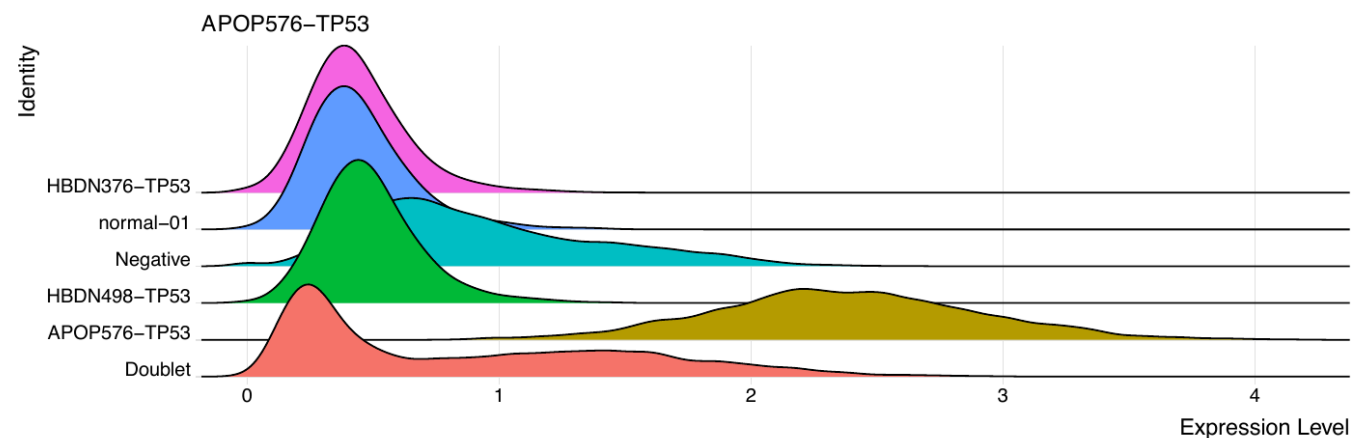
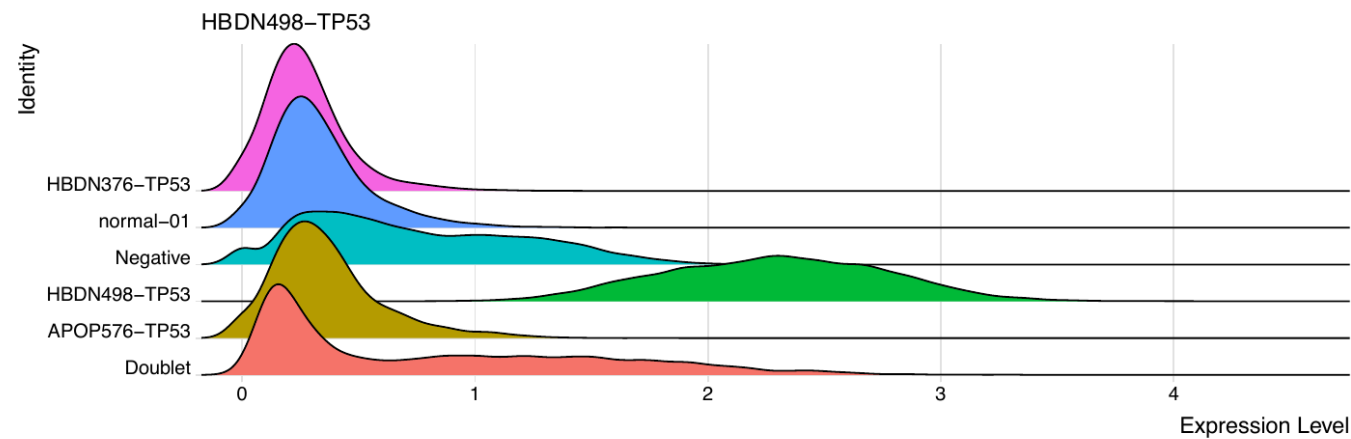
HTO QC at the adopted threshold.

HTO QC; HTODemux positive.quantile = 0.99

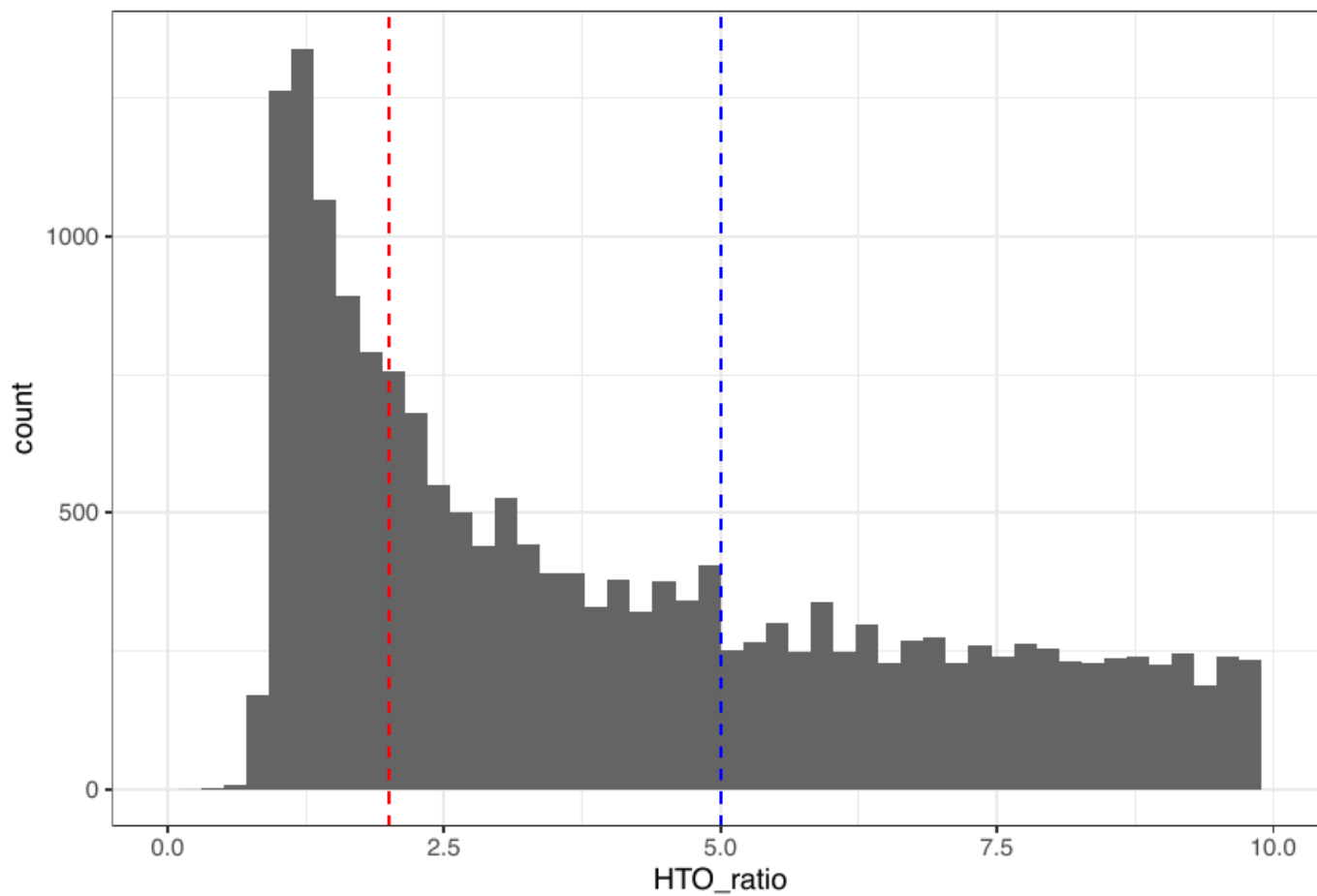


06_HTO_ridge_positive_quantile_0.99.pdf

Ridge plots per hashtag.



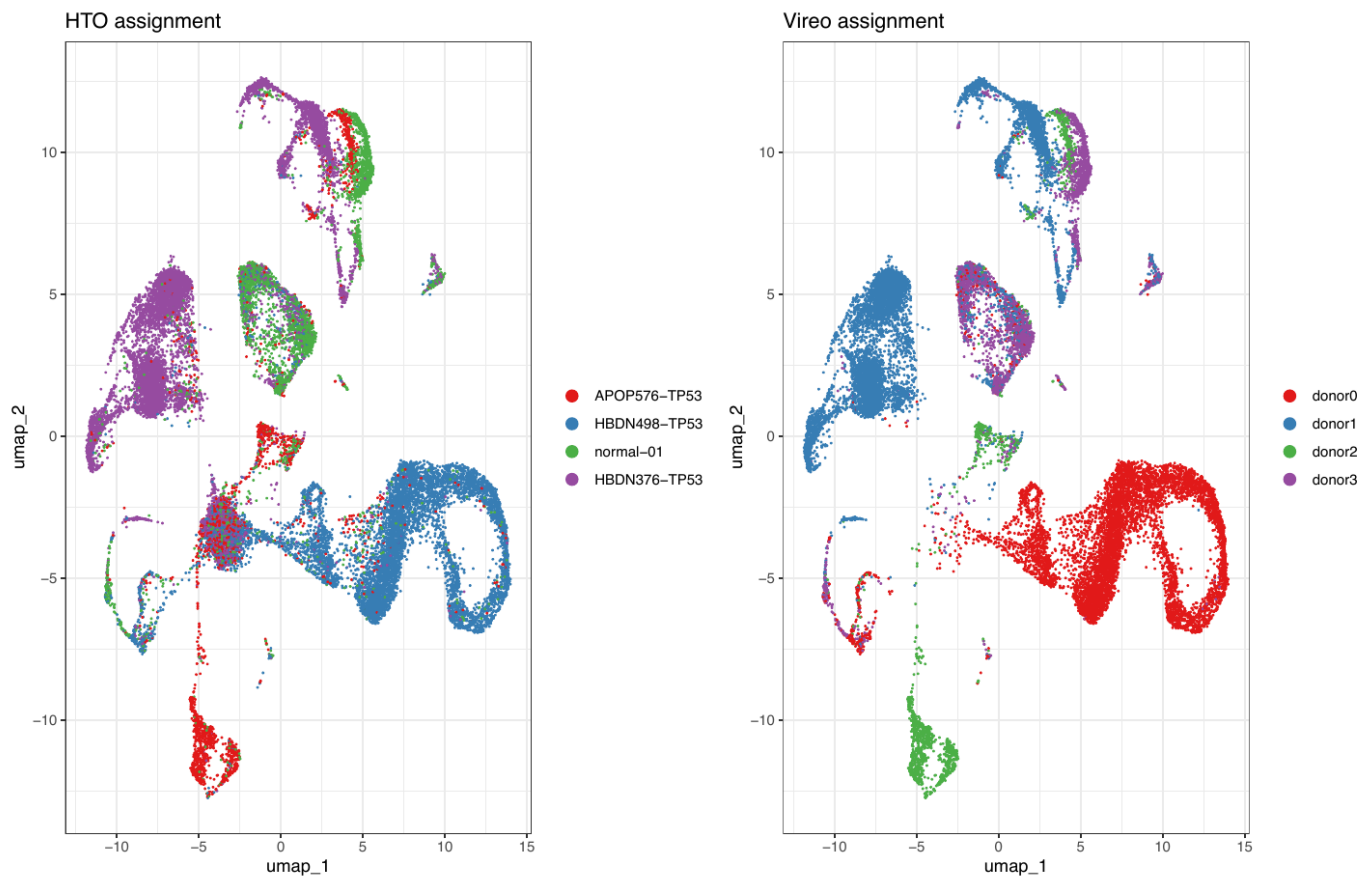
HTO ratio thresholds; HTODemux positive.quantile = 0.99



07_UMAP_HTO_cellSNP_positive_quantile_0.99.pdf

Both sets of calls on the UMAP.

HTO and Vireo assignments; HTODemux positive.quantile = 0.99



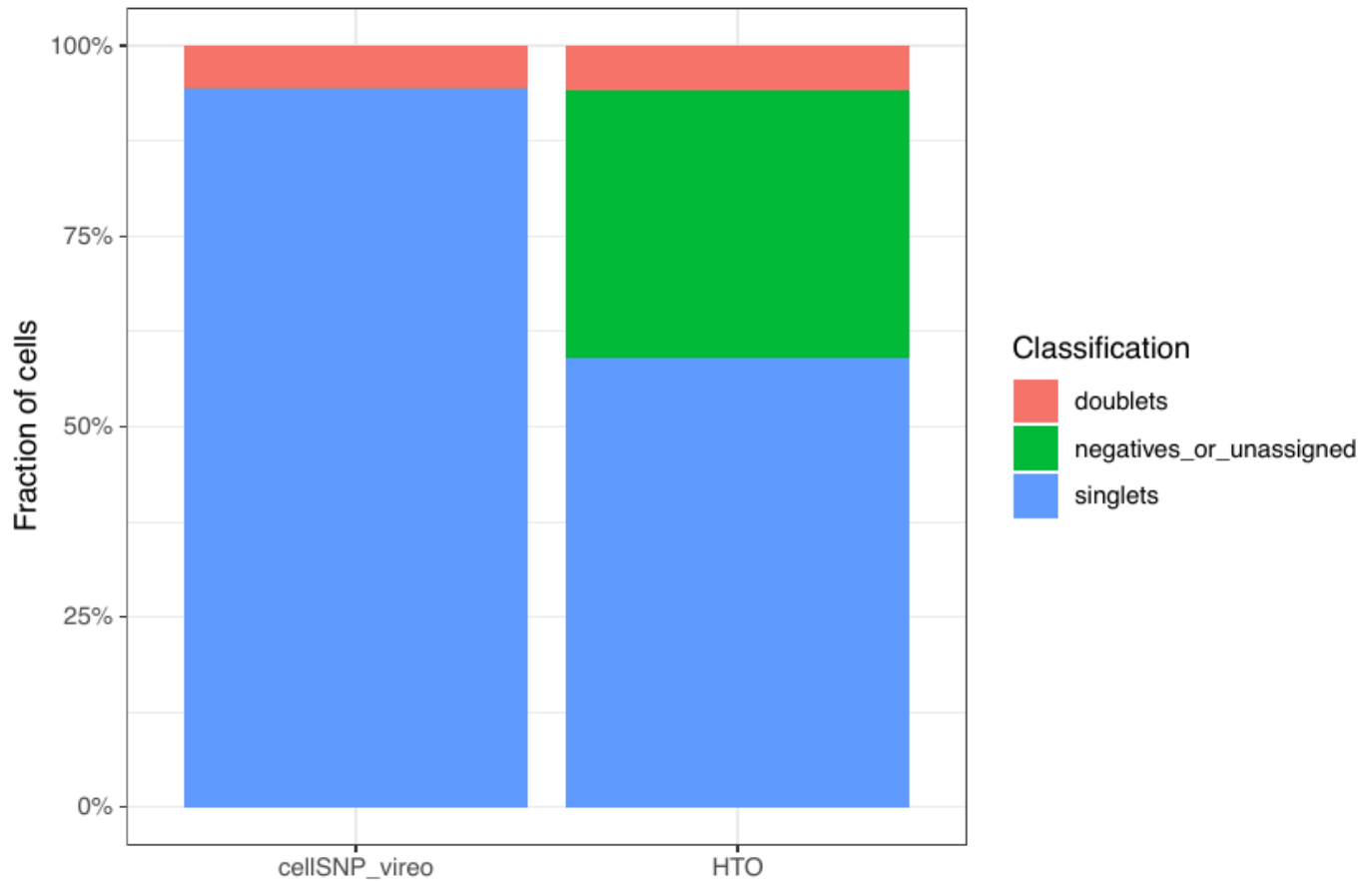
260717

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Singlet / doublet / negative rates for HTO and vireo side by side. The headline comparison.

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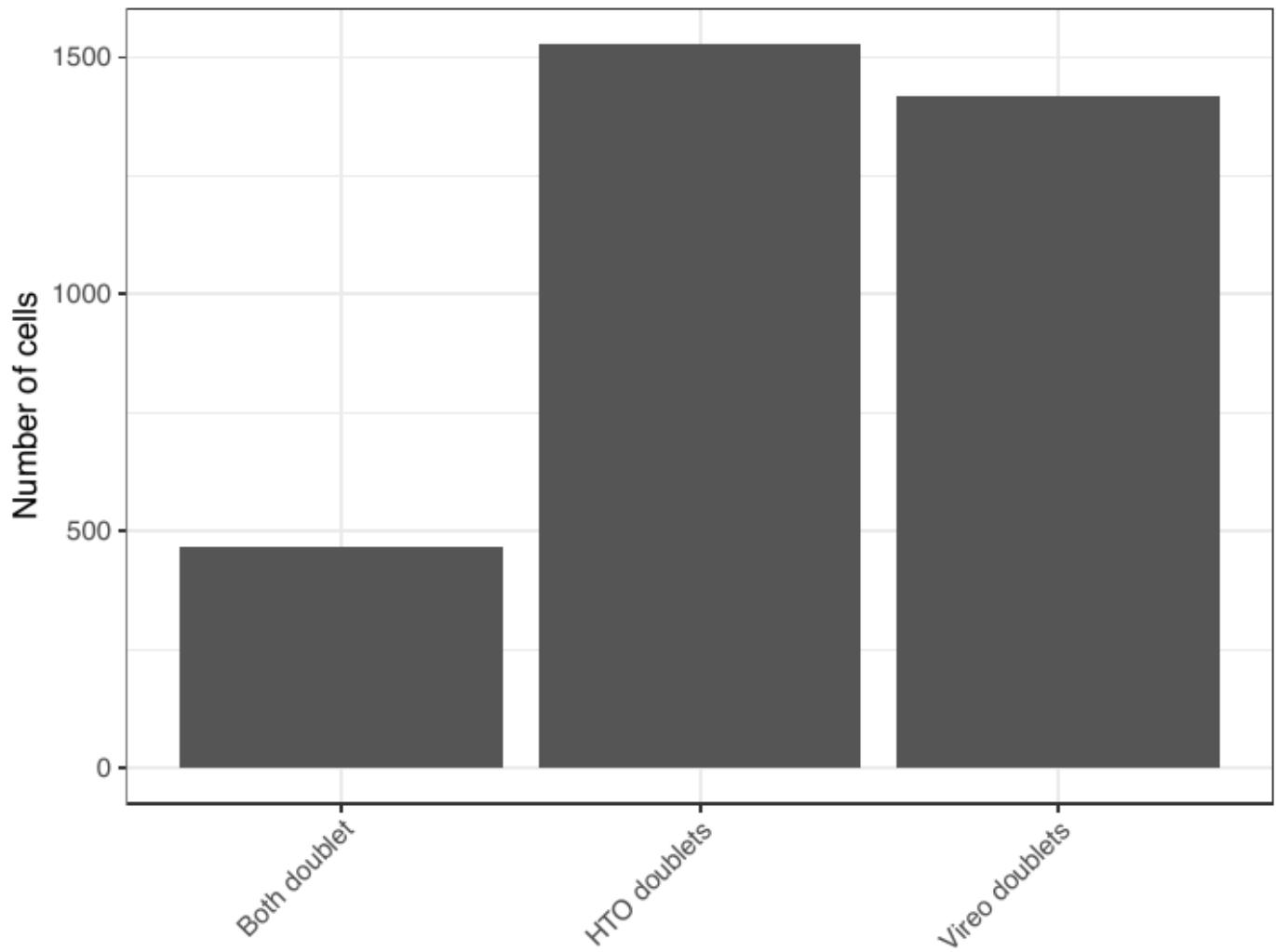
HTODemux positive.quantile = 0.99



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Whether the two methods call the same cells doublets.

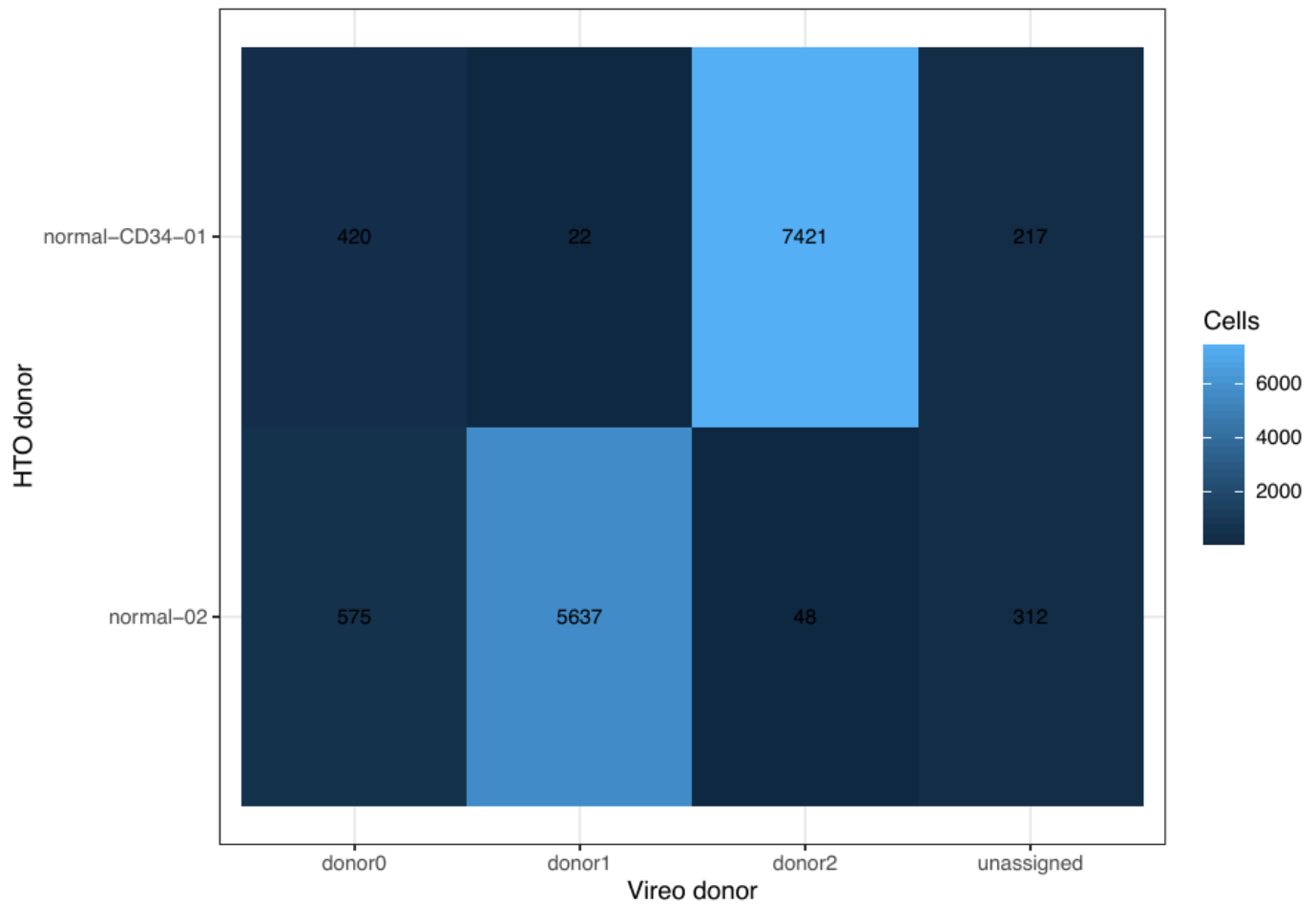
Doublet detection overlap
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03_hto_vireo_donor_heatmap_positive_quantile_0.99.pdf

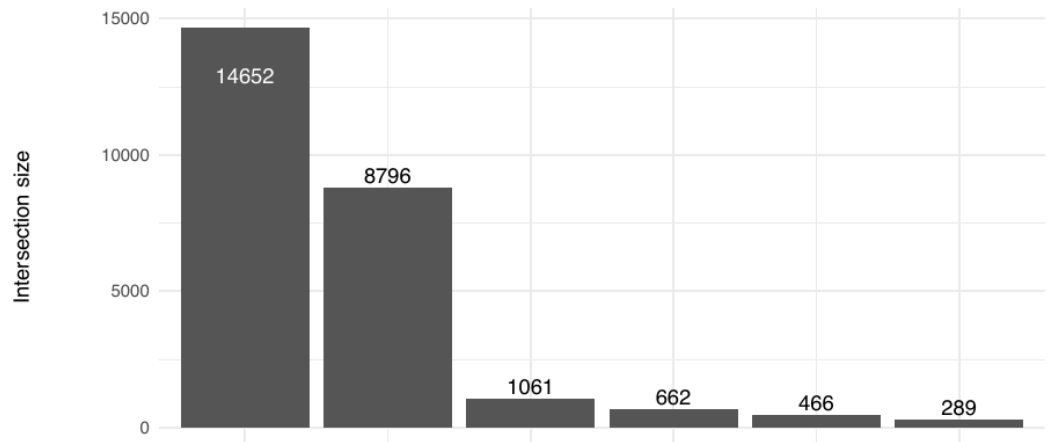
HTO sample against vireo donor. Off-diagonal mass is the disagreement; this is also the figure the sample-name mapping is read off.

HTO versus Vireo donor assignment
HTODemux positive.quantile = 0.99

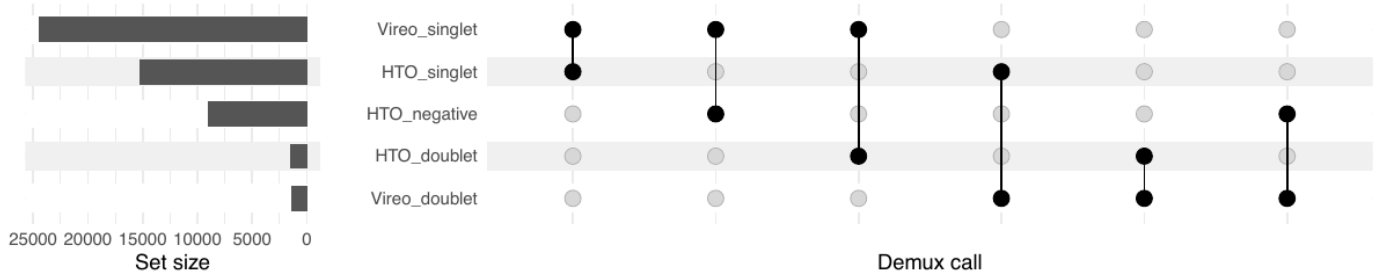


04_upset_HTO_vireo_overlap_positive_quantile_0.99.pdf

UpSet of the call overlap.



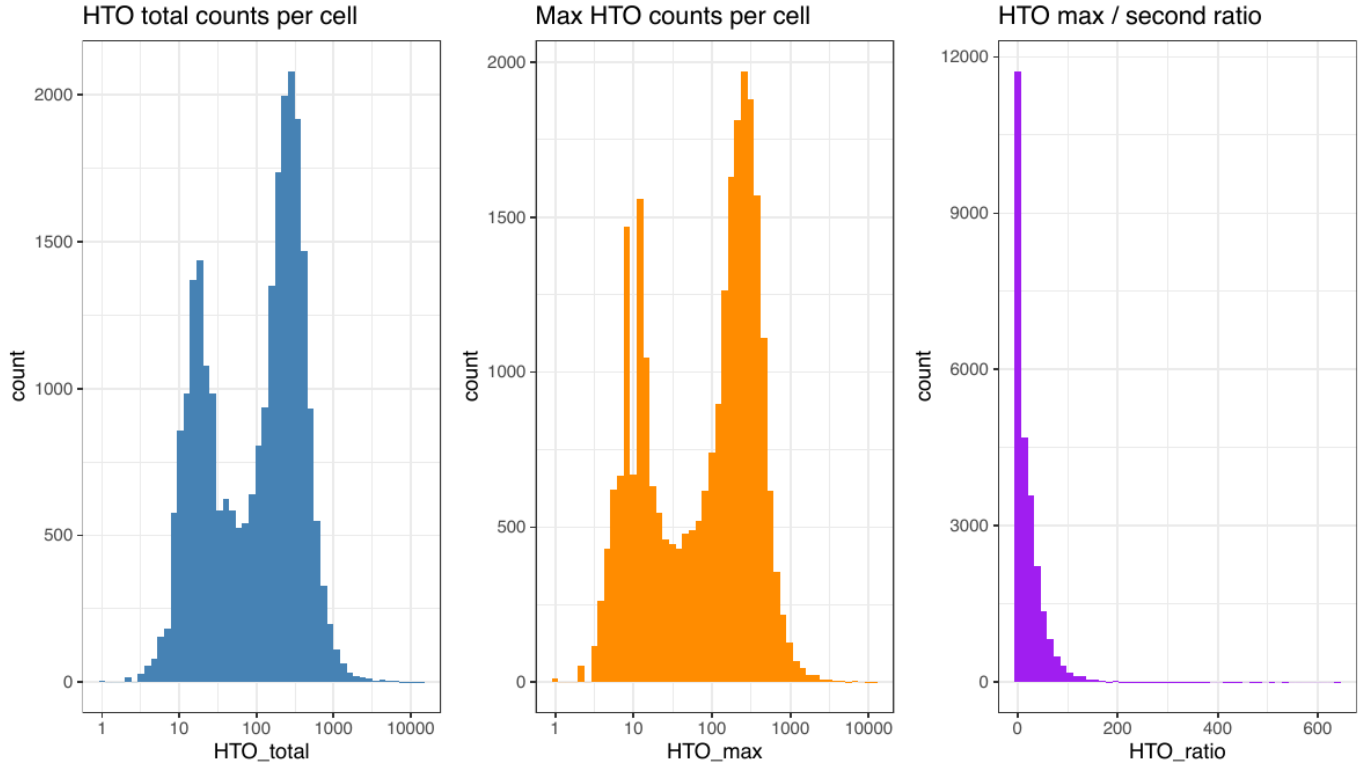
Overlap between HTO and Vireo demultiplexing calls
HTODemux positive.quantile = 0.99



05_HTO_QC_positive_quantile_0.99.pdf

HTO QC at the adopted threshold.

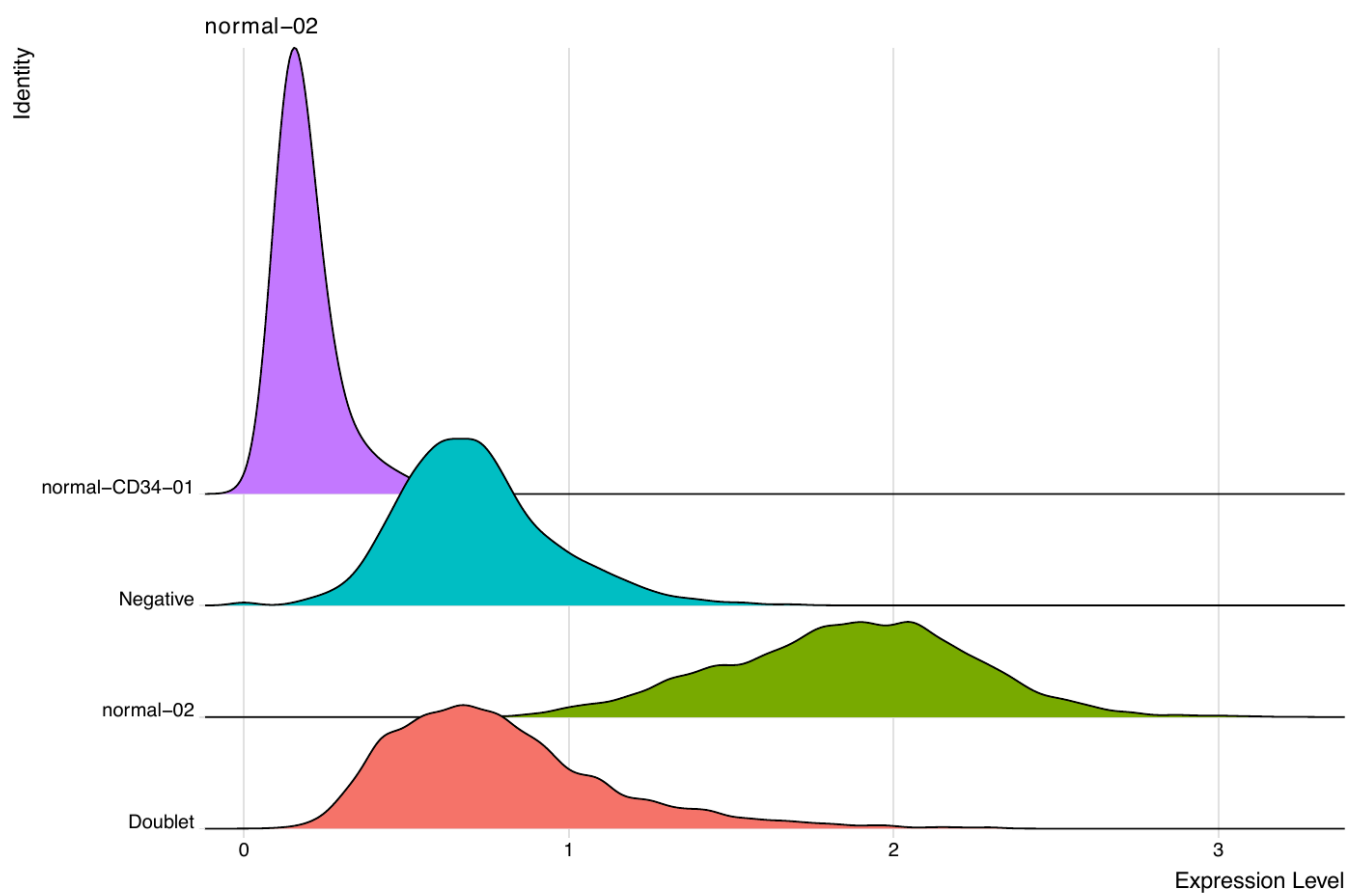
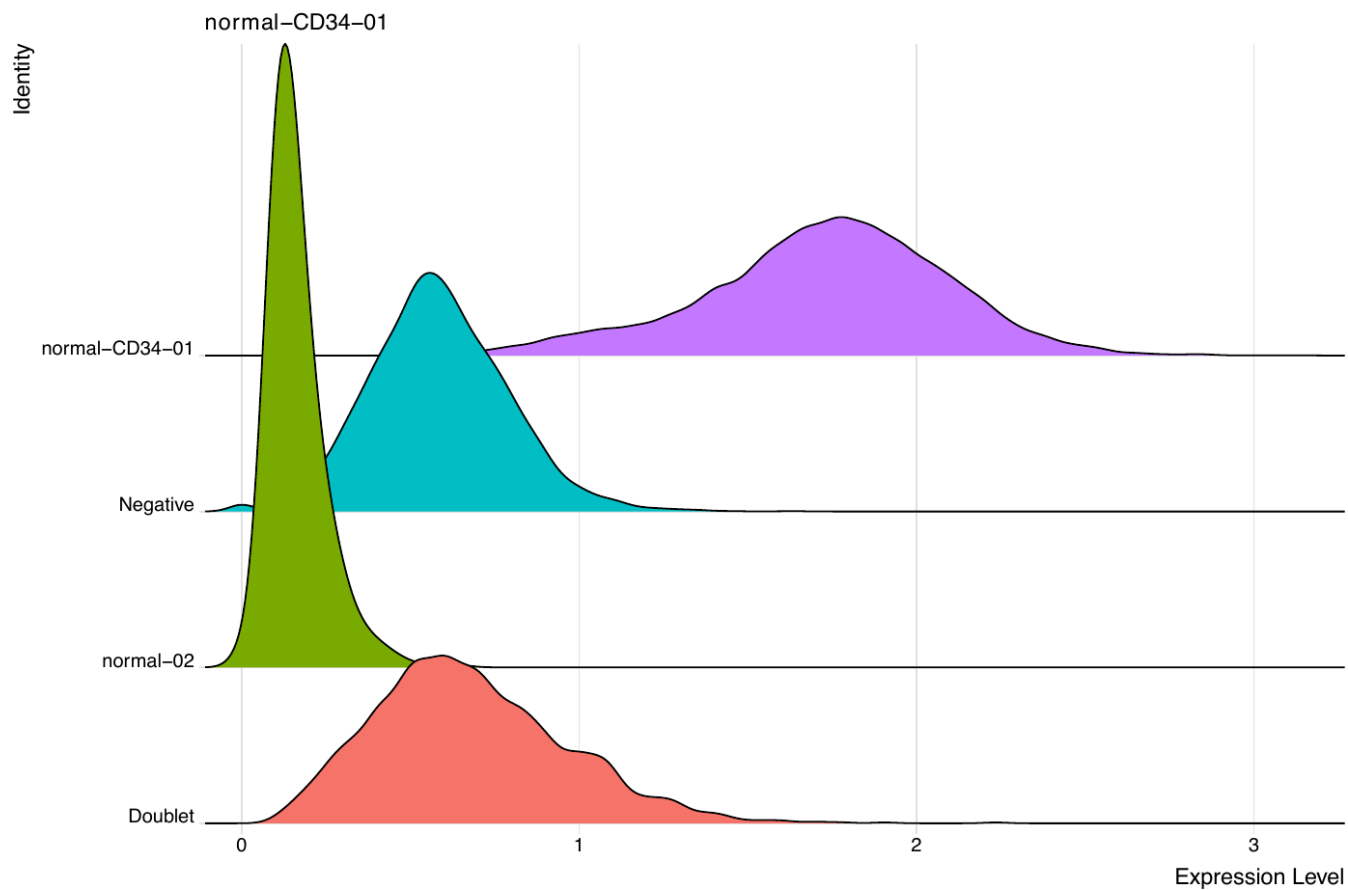
HTO QC; HTODemux positive.quantile = 0.99



06_HTO_ridge_positive_quantile_0.99.pdf

Ridge plots per hashtag.

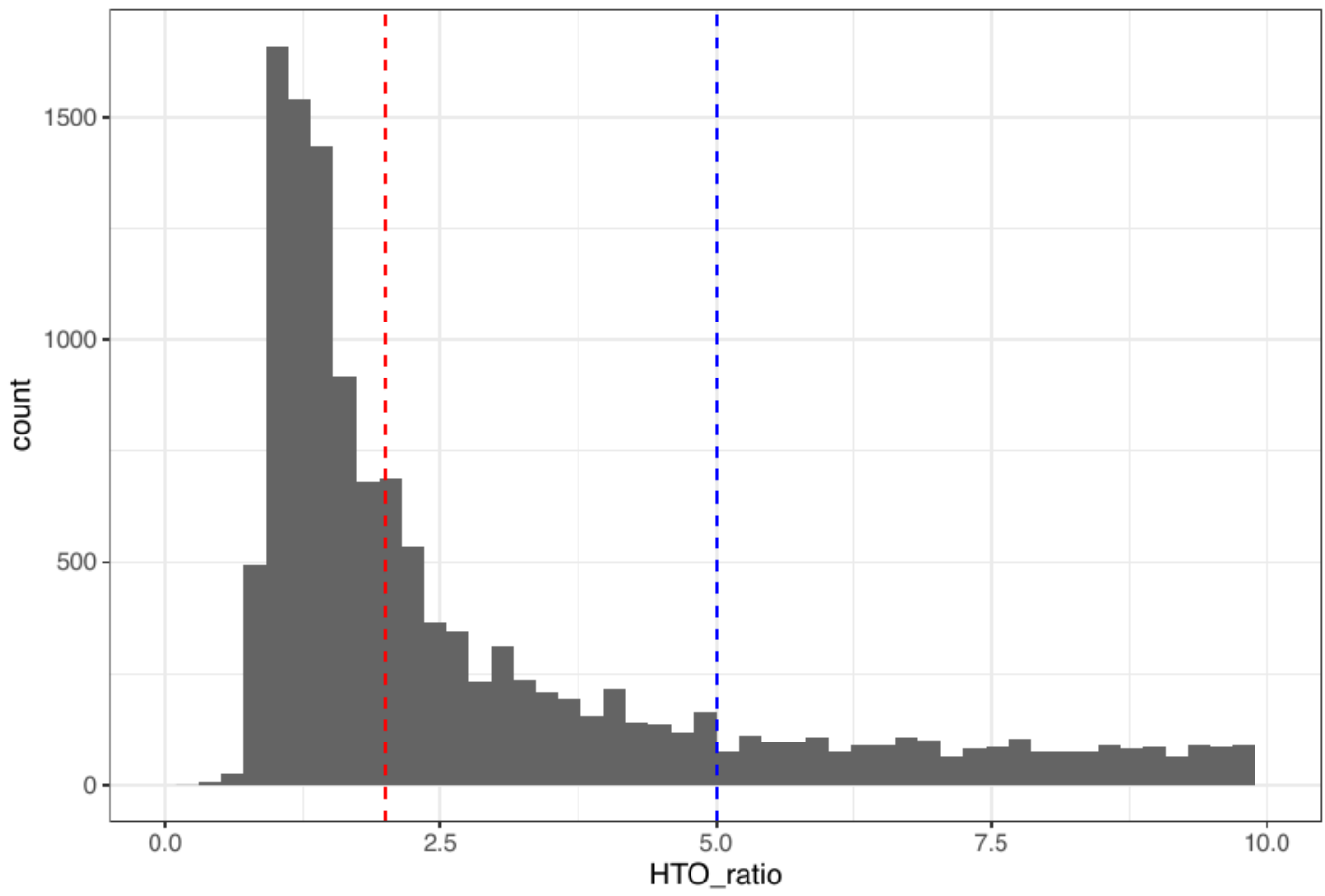
HTO ridge plot; HTODemux positive.quantile = 0.99



06b_HTO_ratio_thresholds_positive_quantile_0.99.pdf

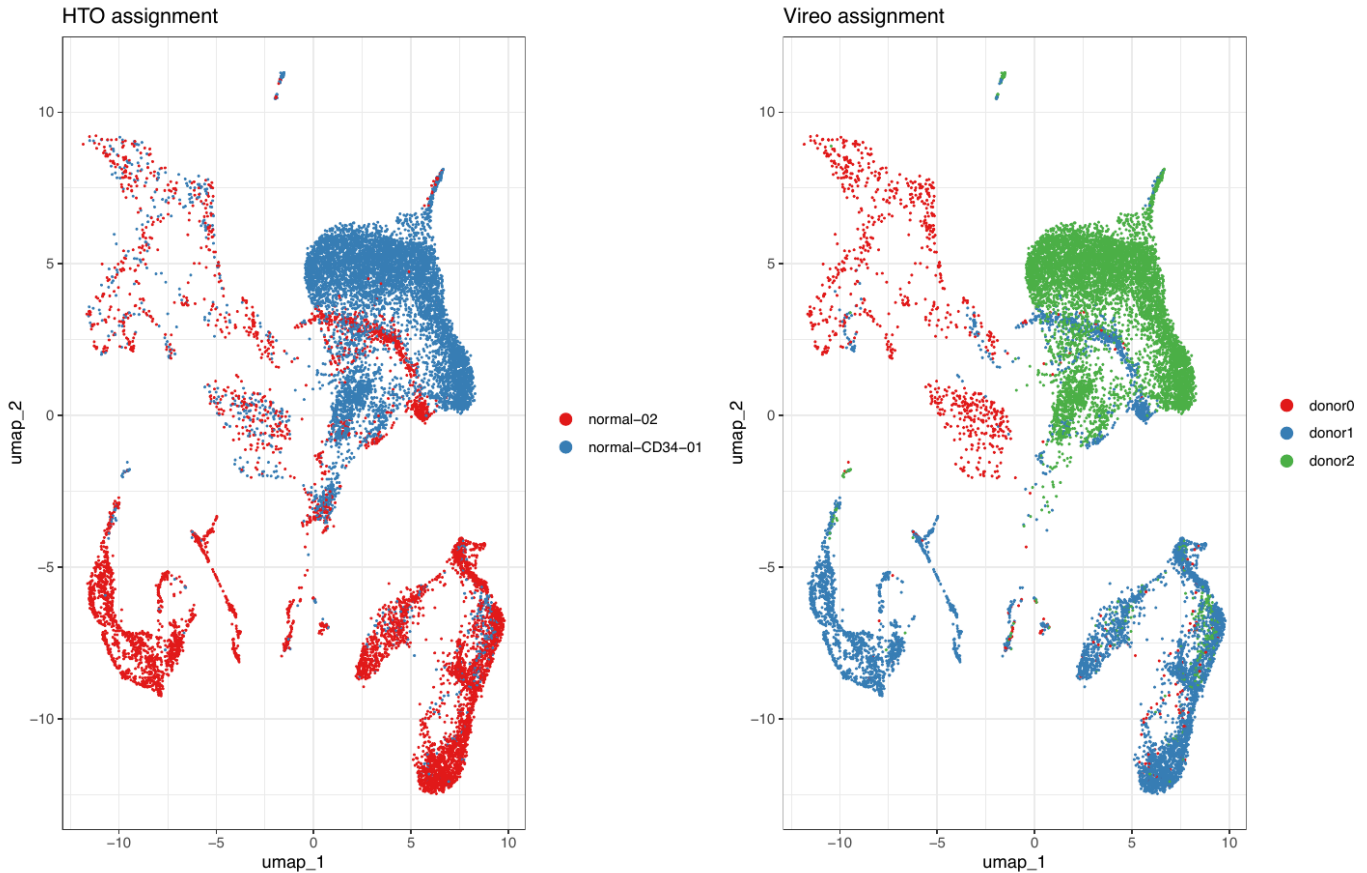
HTO ratio thresholds.

HTO ratio thresholds; HTODemux positive.quantile = 0.99



07_UMAP_HTO_cellSNP_positive_quantile_0.99.pdf

Both sets of calls on the UMAP.



5. The unhashed donor

Not every donor in a pool carries a hashtag. Step 06 checks that case: it runs the production script `nextflow/bin/process_cellsnp_hto.R` with exactly the arguments the pipeline module would pass for LK3-GEX, against that run's already-completed cellSNP/vireo and HTO output. Running the real script rather than a reimplementaion is the point - it exercises the code that ships.

The trial output lives in `scripts/TRIAL_output_unhashed_donor_demux/`, deliberately outside `nextflow/`, following the convention that trial and diagnostic scripts never sit inside the pipeline directory.

6. What was run, in order

Step	Script	What it does
01	<code>01.hto_count.sbatch</code>	CITE-seq-Count on the HTO library, whitelisted to emptyDrops barcodes.
02	<code>02.cellsnp_vireo.sh</code>	cellsnp-lite pileup over the same barcodes, then vireo with <code>N_DONORS=4</code> .
03	<code>03.process_cellsnp_HTO.R</code>	Joins vireo onto the object, cross-tabulates against HTO, writes the sweep.
04	<code>04.QC_HTO_ADT_only.R</code>	QC restricted to the HTO and ADT modalities.
05	<code>05.QC_HTO_comparison.R</code>	Compares HTO and ADT behaviour between the LK1 and LK2 runs.
06	<code>06.test_unhashed_donor_demux.sh</code>	Verifies the unhashed-donor fix against LK3's real output.
07	<code>07.demux_benchmark_report.Rmd</code>	This report.

The biotin-specific part of step 05 was later split into `scripts/6b1.biotin_ADT_comparison.R`, which belongs to the biotin report rather than here.

7. Data locations

Demux comparison outputs by run and source

What	Location
Genotype demux	results/genotype_demux/<run>/<sample>/cellsnp/ and ../vireo/
Comparison, standalone	results/demux_comparison/<run>/<sample>/
Comparison, pipeline	results_nf/<run>/06_demux_comparison/
HTO counts	results/cite_seq_count/<run>/hto_counts_*/
Unhashed-donor trial	scripts/TRIAL_output_unhashed_donor_demux/

run	source	files	first	last
260423	standalone	32	2026-05-04	2026-05-05
260522	pipeline	18	2026-07-28	2026-07-28
260528	standalone	24	2026-06-02	2026-06-03
260528	pipeline	18	2026-07-27	2026-07-27
260717	pipeline	18	2026-07-21	2026-07-21

8. Tools

Tool	Version	Source
cellsnp-lite	1.2.3 (htslib 1.23.1)	cellsnp_env
vireo (vireoSNP)	to confirm	cellsnp_env
CITE-seq-Count	to confirm	01.hto_count.sbatch
Seurat HT0Demux	see session below	positive.quantile = 0.99 adopted

```
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1
##
## locale:
## [1] en_AU/en_AU/en_AU/C/en_AU/en_AU
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] scales_1.4.0  ggplot2_4.0.2 knitr_1.51   purrr_1.2.1  stringr_1.6.0
## [6] readr_2.1.6   tidyr_1.3.2   dplyr_1.2.0
##
## loaded via a namespace (and not attached):
## [1] bit_4.6.0      gtable_0.3.6   jsonlite_2.0.0  crayon_1.5.3
## [5] qpdf_1.4.1     compiler_4.5.1 pdftools_3.9.0  Rcpp_1.1.1
## [9] tidyselect_1.2.1 parallel_4.5.1 jquerylib_0.1.4  yaml_2.3.12
## [13] fastmap_1.2.0  R6_2.6.1       generics_0.1.4  tibble_3.3.1
## [17] RColorBrewer_1.1-3 bslib_0.10.0   pillar_1.11.1   tzdb_0.5.0
## [21] rlang_1.1.7    cachem_1.1.0   stringi_1.8.7    xfun_0.56
## [25] S7_0.2.1       sass_0.4.10    bit64_4.6.0-1    otel_0.2.0
## [29] cli_3.6.5      withr_3.0.2    magrittr_2.0.4   digest_0.6.39
## [33] grid_4.5.1     vroom_1.7.0    askpass_1.2.1    hms_1.1.4
## [37] lifecycle_1.0.5 vctrs_0.7.1    evaluate_1.0.5   glue_1.8.0
## [41] farver_2.1.2   rmarkdown_2.30 tools_4.5.1      pkgconfig_2.0.3
## [45] htmltools_0.5.9
```

9. Caveats

1. **The sweep exists only for the pilot run.** From LK2 onwards only the 0.99 outputs were generated, so the comparison across thresholds cannot be redone from saved files for the later runs without re-running step 03.

2. **results/demux_comparison/** has no entry for LK1-resequenced or LK3. Those runs were demultiplexed through the pipeline, so their comparison lives under `results_nf/<run>/`.
3. Vireo's donor labels are anonymous; any statement tying a genotype cluster to a sample name depends on the HTO mapping in `vireo_to_hto_sample_name_map_positive_quantile_0.99.csv`.